

ECEN 607: Advanced Analog Circuit Tech

Lab 1: MOS Device Characterization Lab Report

Name: Sheena Selvakumar

UIN: 537000654

Texas A&M University ECEN 607: Adv. Analog Circuits & Tech Spring 2026

Objectives:

To review concepts learned in background classes such as characterization, small-signal and CMOS circuit design.

5 V Transistor Characterization

High voltage CMOS design is done in many applications even today. In this course, you will be designing circuits operating both at 1.8 V supply and 5.5 V supply. Using the technology process IBM180 and the following procedure, characterize the four terminal, **5.5 V** nominal, MOS transistors **nfetmx** and **pfetmx**.

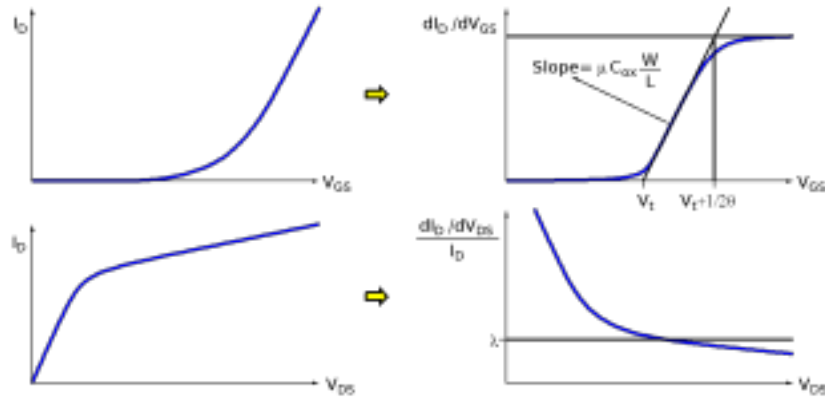
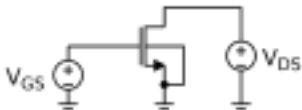


Figure 1-11: Curves used for CMOS characterization

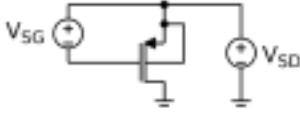
To obtain g_m , derivative of I_D needs to be plotted as a function of V_{GS} . The resulting plot can be used to extract V_t , μC_{ox} and θ , as shown above. Channel length modulation parameter lead to the finite value of the output conductance, and can be extracted by taking the derivative of I_D as a function of V_{DS} .

1. Construct the circuit below using **nfetmx** transistor with $W/L=1\mu/0.7\mu$.



- a. Fix $V_{DS}=2V$. Sweep V_{GS} from 0 to 5V, plot I_D and g_m (dI_D/dV_{GS}) vs. V_{GS} , and extract V_{tn} , $\mu_n C_{ox}$ and θ .
- b. Set $V_{GS}=V_{tn}+100mV$. Sweep V_{DS} from 0 to 5V, plot and g_{ds} (dI_D/dV_{DS}) vs. V_{DS} . Plot the output resistance r_{ds} , and its first and second derivative as function of V_{DS} .
- c. Find the intrinsic gain (A_i) and unity-gain frequency (f_t) using AC simulations when $V_{DS}=2V$ and $V_{GS}=V_{tn}+100mV$.

2. Repeat (1) for the **nfetmx** transistor with $W/L=5\mu/0.7\mu$.
3. Repeat (1) for the **nfetmx** transistor with $W/L=5\mu/1.4\mu$.
4. Construct the circuit below using **pfetmx** transistor with $W/L=1\mu/0.5\mu$.



- a. $V_{SD}=2V$. Sweep V_{SG} from 0 to 5V, plot I_D and $gm=dI_D/dV_{SG}$ vs. V_{SG} , and extract V_{tp} , $\mu_p C_{ox}$ and θ .
- b. $V_{SG}=|V_{tp}|+100mV$. Sweep V_{SD} from 0 to 5V, plot and $g_{ds}(dI_D/dV_{SD})$ vs. V_{DS} . Plot the output resistance r_{ds} , and its first and second derivative as function of V_{SD} .
- c. Find the intrinsic gain (A_i) and unity-gain frequency (f_t) using AC simulations when $V_{SD}=2V$ and $V_{SG}=|V_{tp}|+100mV$.
5. Repeat (4) for the **pfetmx** transistor with $W/L=5\mu/0.5\mu$.
6. Repeat (4) for the **pfetmx** transistor with $W/L=5\mu/1\mu$.

In the lab report, include your schematics, plots, annotations, and calculations. Please provide the characterization results in a plots and comment on the results.

DISCUSSION OF THE LAB AND RESULTS

- The primary objective of this lab was to characterize key process parameters of NMOS and PMOS transistors, such as threshold voltage (V_{th}), body effect coefficient (θ), channel length modulation parameter (λ), unity gain frequency (f_T), and intrinsic gain (A_i), through both simulation data and analytical extraction.
- Pre-lab designs were developed based on theoretical calculations and expected process characteristics, serving as an initial benchmark.
- For θ extraction, three pairs of (v_t , $v_t + 1/(2\theta)$) data points were analyzed, yielding θ values between approximately 0.33 and 0.59. These values varied based on device geometry and operating bias and were generally in good agreement with pre-lab expectations, necessitating only minor adjustments.
- Such adjustments highlight the limitations of analytical models versus real device behavior, underscoring the importance of simulation and model refinement.

- Simulation trends observed included:
 - Short-channel devices displayed higher θ values due to pronounced channel length modulation.
 -
- The lab demonstrated key design trade-offs, such as balancing device speed (fT), gain (Ai), and power consumption.
- In rare cases where specifications were not fully met, the main contributing factors were:
 - Omission of non-ideal effects in pre-lab calculations,
 - Limitations in device sizing during the design phase.

II. SUMMARY OF RESULTS

Summarize the simulation results from the lab in one or two tables, like what is shown below.

Table 1

Example summary of results for process parameter characterization -nmos

Parameter	$\mu C_{ox} W/L$	V_{th}	θ	λ	f_T	A_i
$W/L=1\mu/0.7\mu$	169.5u	507.2mV	0.5412	214.0730n	4.5022GHz	38.2774db
$W/L=5\mu/1.4\mu$	926.8u	505.6mV	0.5866	1.1328u	5.19171GHz	38.438db
$W/L=1\mu/0.5\mu$	371.8u	579.6mV	0.3323	99.80n	33.55GHz	52.2094db

Example summary of results for process parameter characterization -pmos

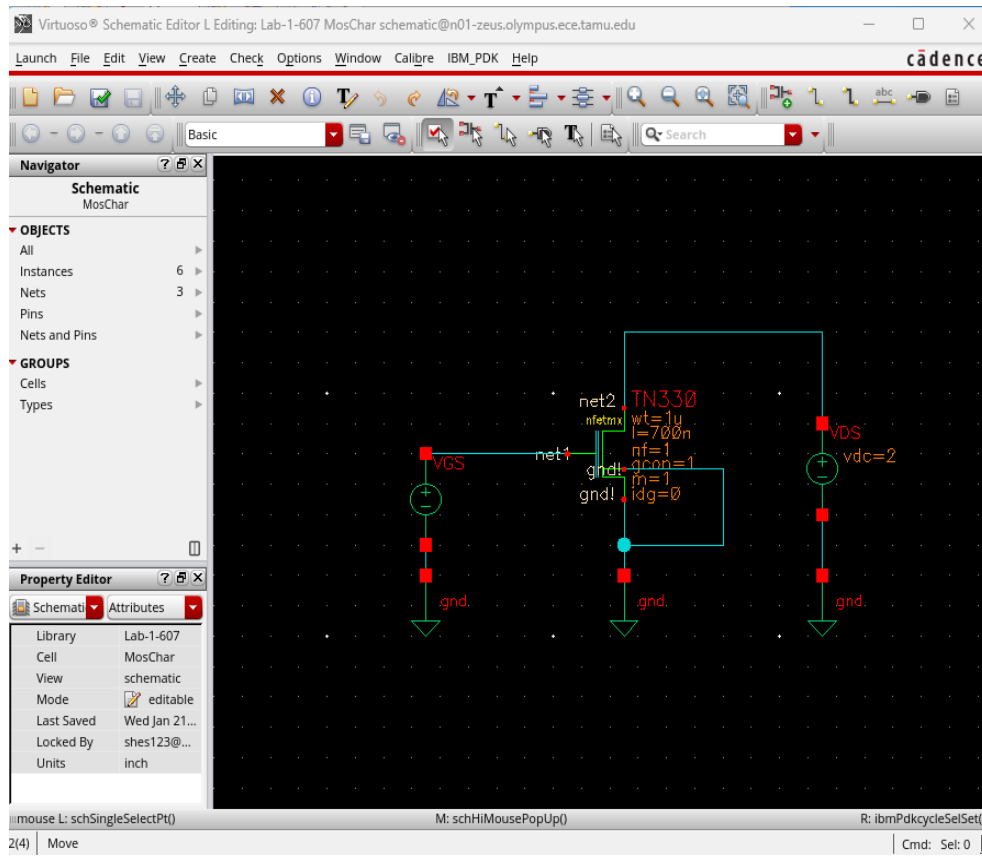
Parameter	$\mu C_{ox} W/L$	V_{th}	θ	λ	f_T	A_i
$W/L=1\mu/0.5\mu$	-60.63u	478.2mV	0.5001	218.642n	2.81497GHz	33.8987db
$W/L=5\mu/0.5\mu$	-357.4u	473.7mV	0.4478	740.232n	3.46855GHz	33.9949db
$W/L=5\mu/1\mu$	-126.6u	592.9mV	0.4999	47.44n	910.371GHz	49.327db

III. SUB-CIRCUIT AND TEST BENCH SCHEMATICS

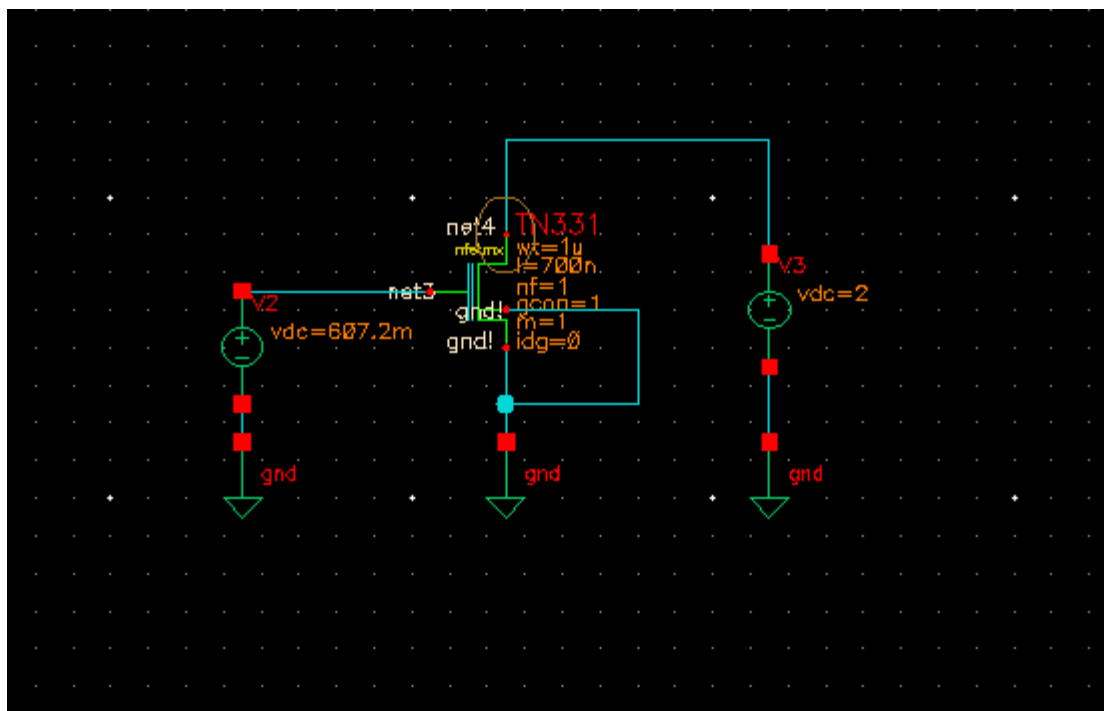
Schematics of the sub-circuit(s) being designed in the lab should be included here. In addition, include images of the test bench schematics. Important signals in the test bench schematic should be labeled to provide clear correspondence between the signals in the schematics and the signals that are plotted in the simulation results shown later in the report.

1. $W/L=1\mu/0.7\mu$:

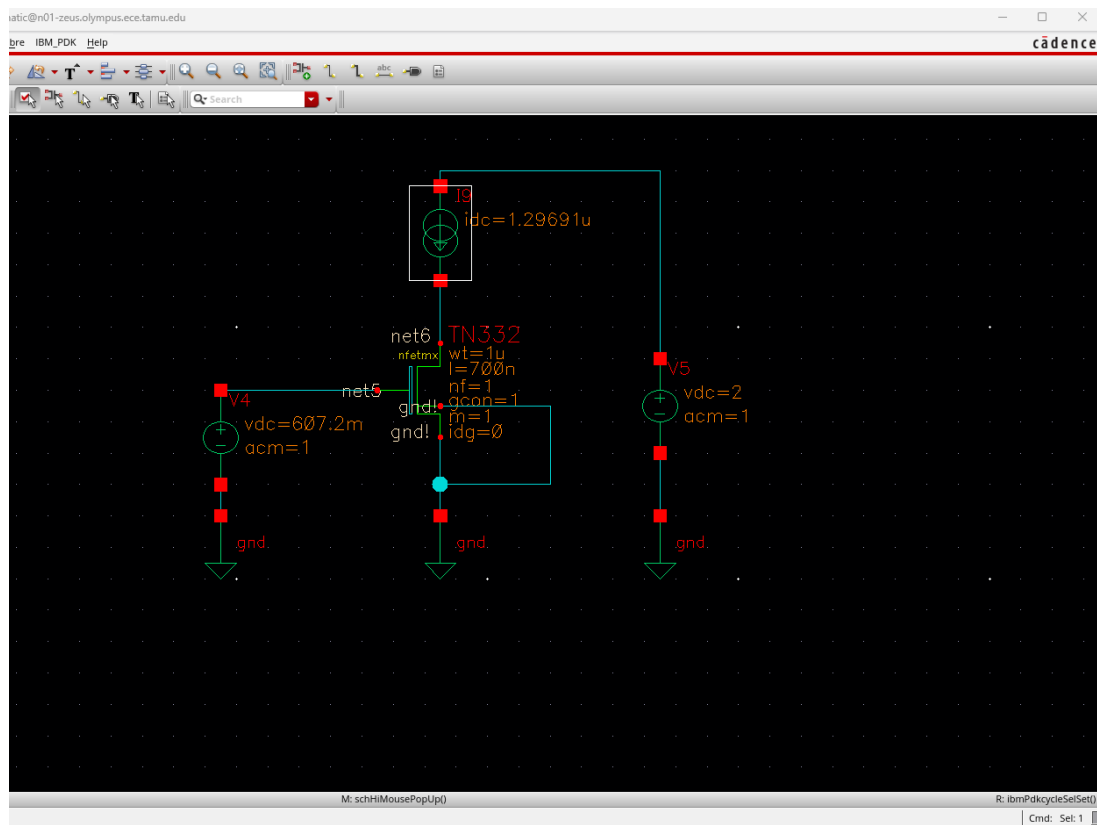
a. Fix $V_{DS}=2V$. Sweep V_{GS} from 0 to 5V, plot I_D and gm (dI_D/dV_{GS}) vs. V_{GS} , and extract V_{tn} , $\mu_n C_{ox}$ and θ .



b. Set $V_{GS} = V_{tn} + 100mV$. Sweep V_{DS} from 0 to 5V, plot and g_{ds} (dI_D/dV_{DS}) vs. V_{DS} . Plot the output resistance r_{ds} , and its first and second derivative as function of V_{DS} .

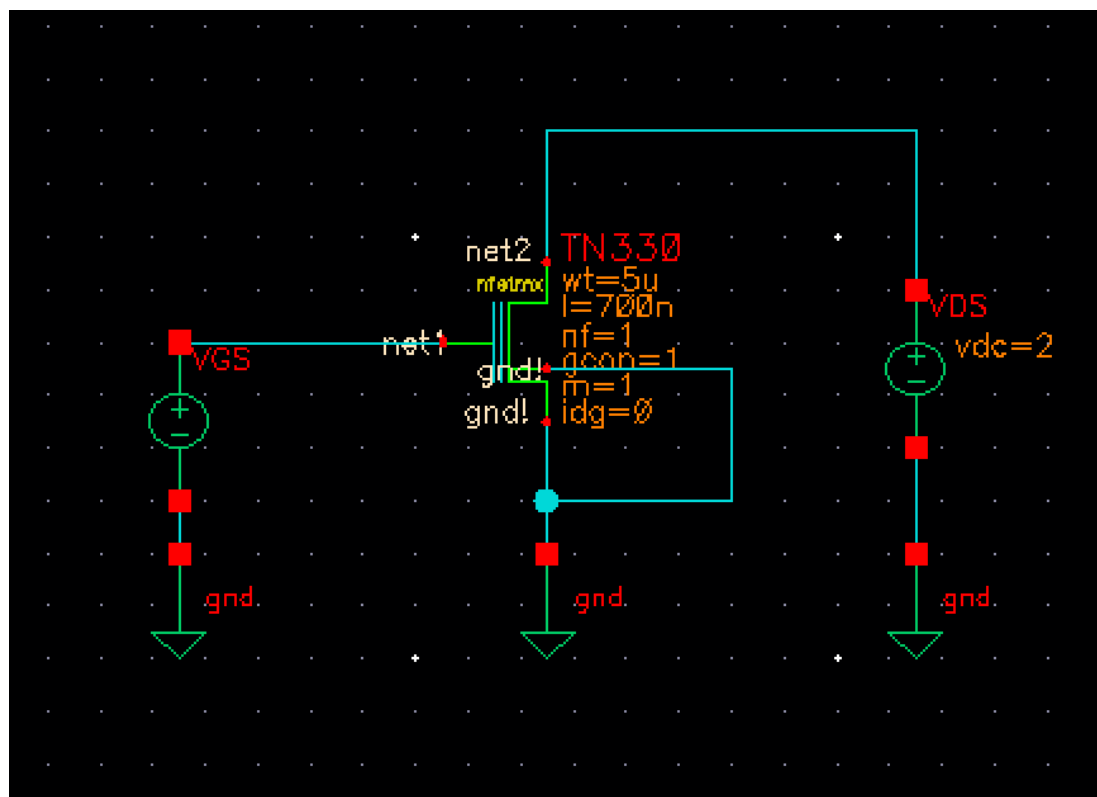


c. Find the intrinsic gain (A_i) and unity-gain frequency (f_t) using AC simulations when $V_{DS} = 2V$ and $V_{GS} = V_{tn} + 100mV$.

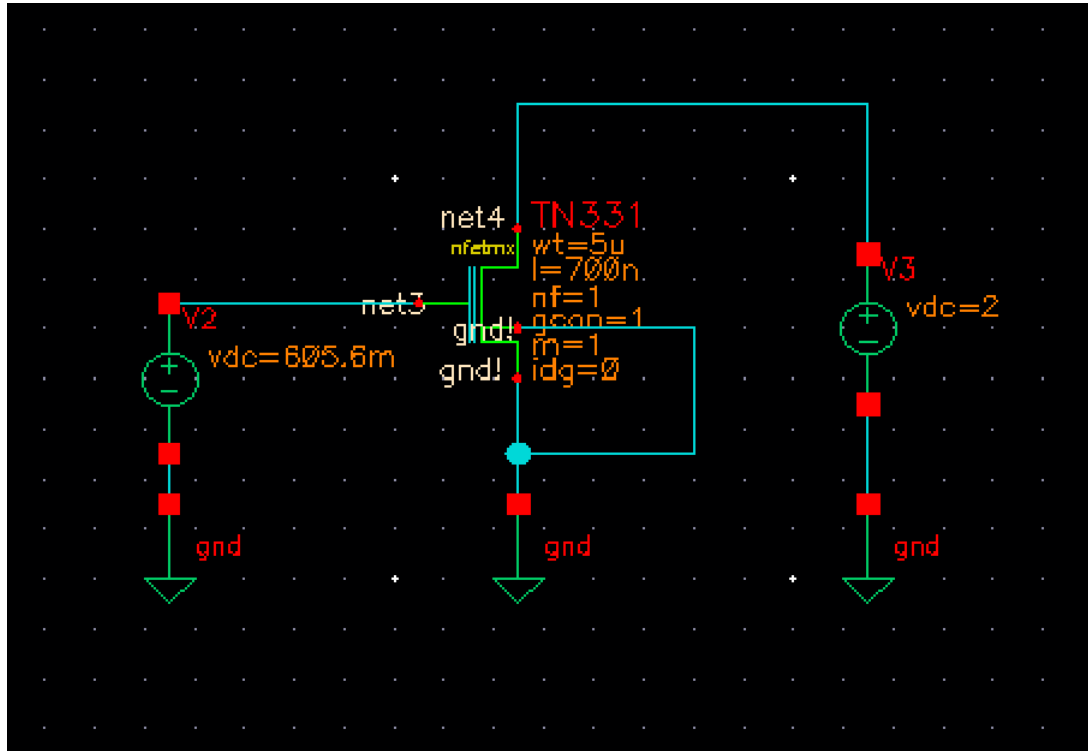


$$2. W/L = 5\mu/0.7\mu$$

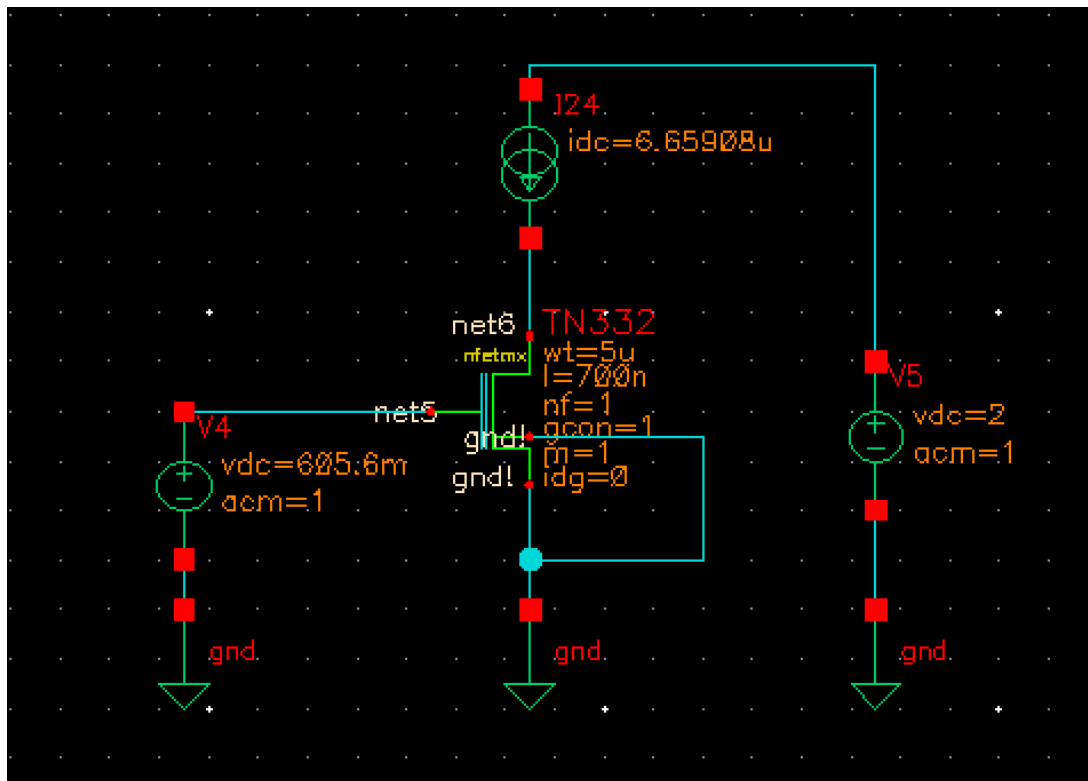
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b. Set $V_{GS}=V_{tn}+100\text{mV}$. Sweep V_{DS} from 0 to 5V, plot and g_{ds} (dI_D/dV_{DS}) vs. V_{DS} . Plot the output resistance r_{ds} , and its first and second derivative as function of V_{DS} .

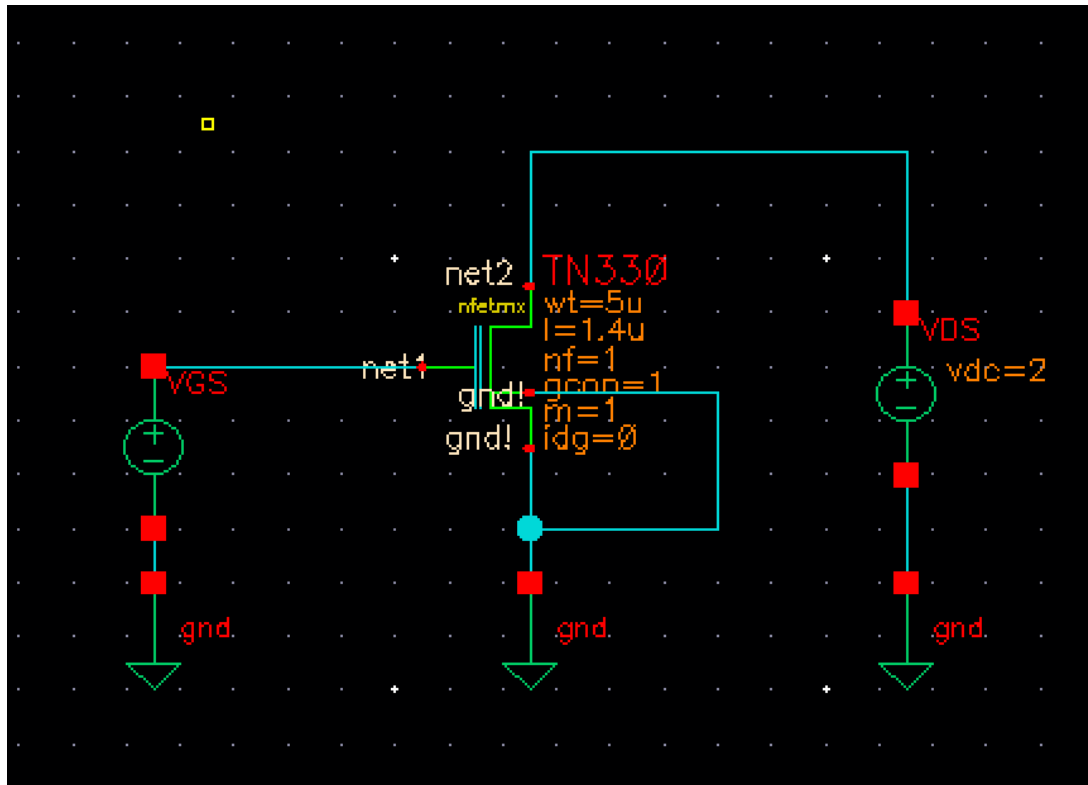


c. Find the intrinsic gain (A_i) and unity-gain frequency (f_t) using AC simulations when $V_{DS}=2\text{V}$ and $V_{GS}=V_{tn}+100\text{mV}$.

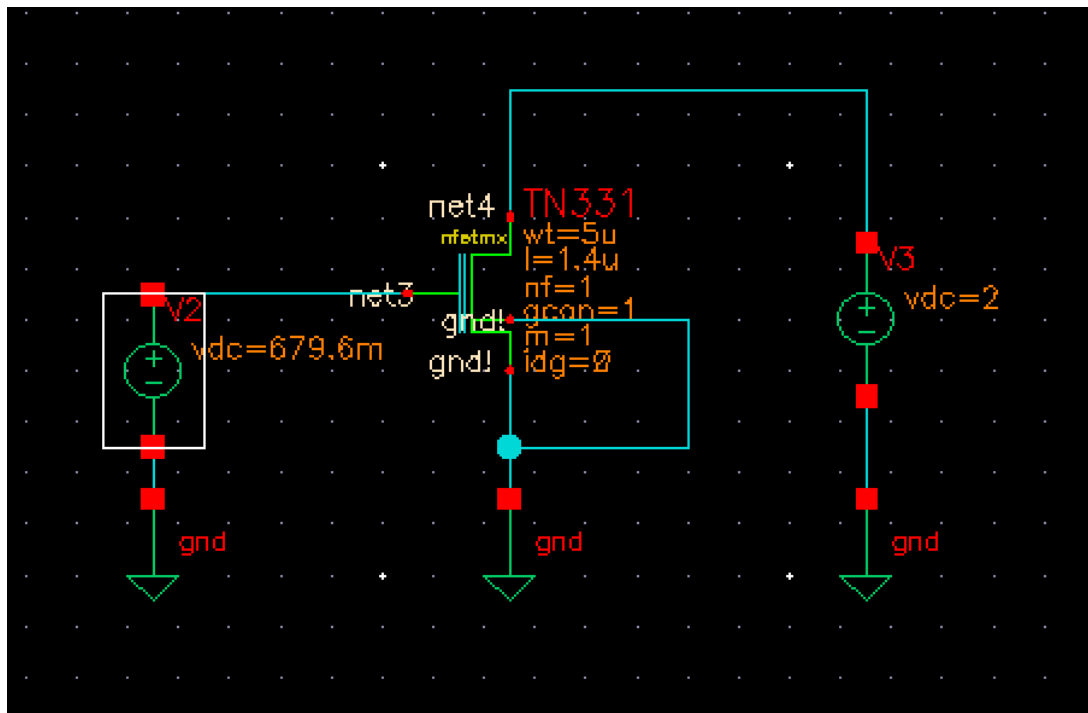


3. $W/L=5\mu/1.4\mu$.

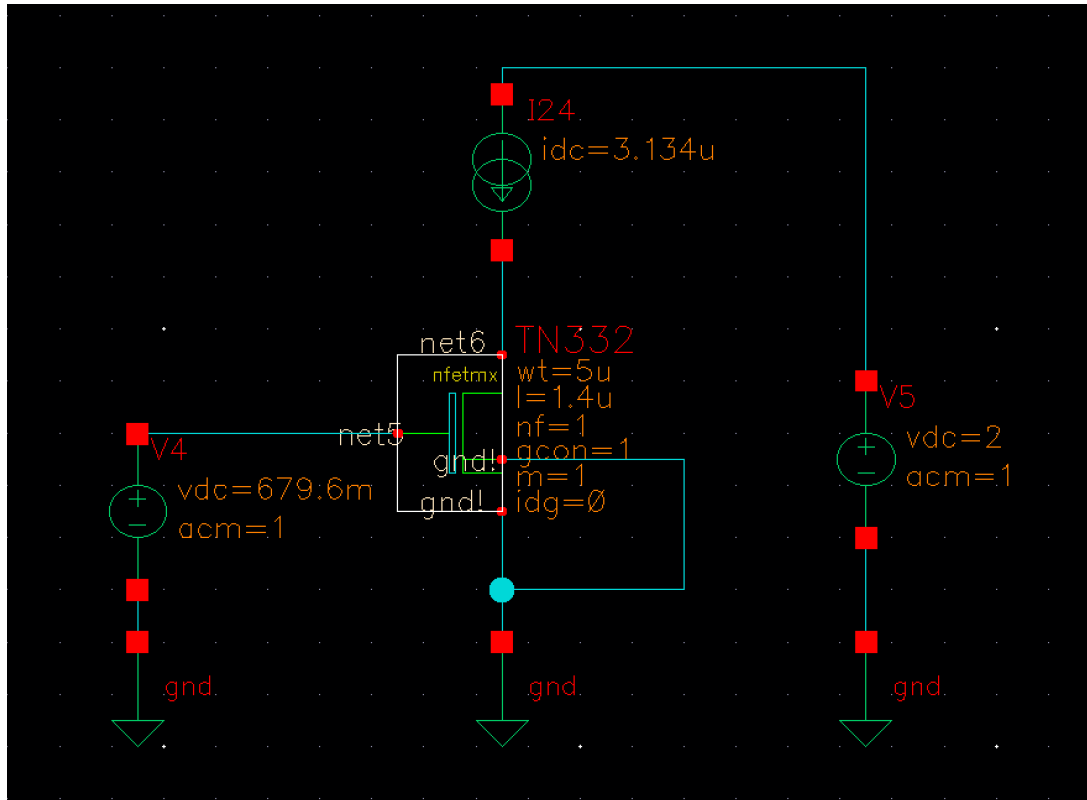
a. Fix $V_{DS}=2V$. Sweep V_{GS} from 0 to 5V, plot I_D and g_m (dI_D/dV_{GS}) vs. V_{GS} , and extract V_{tn} , $\mu_n C_{ox}$ and θ .



b. Set $V_{GS}=V_{tn}+100mV$. Sweep V_{DS} from 0 to 5V, plot and g_{ds} (dI_D/dV_{DS}) vs. V_{DS} . Plot the output resistance r_{ds} , and its first and second derivative as function of V_{DS} .



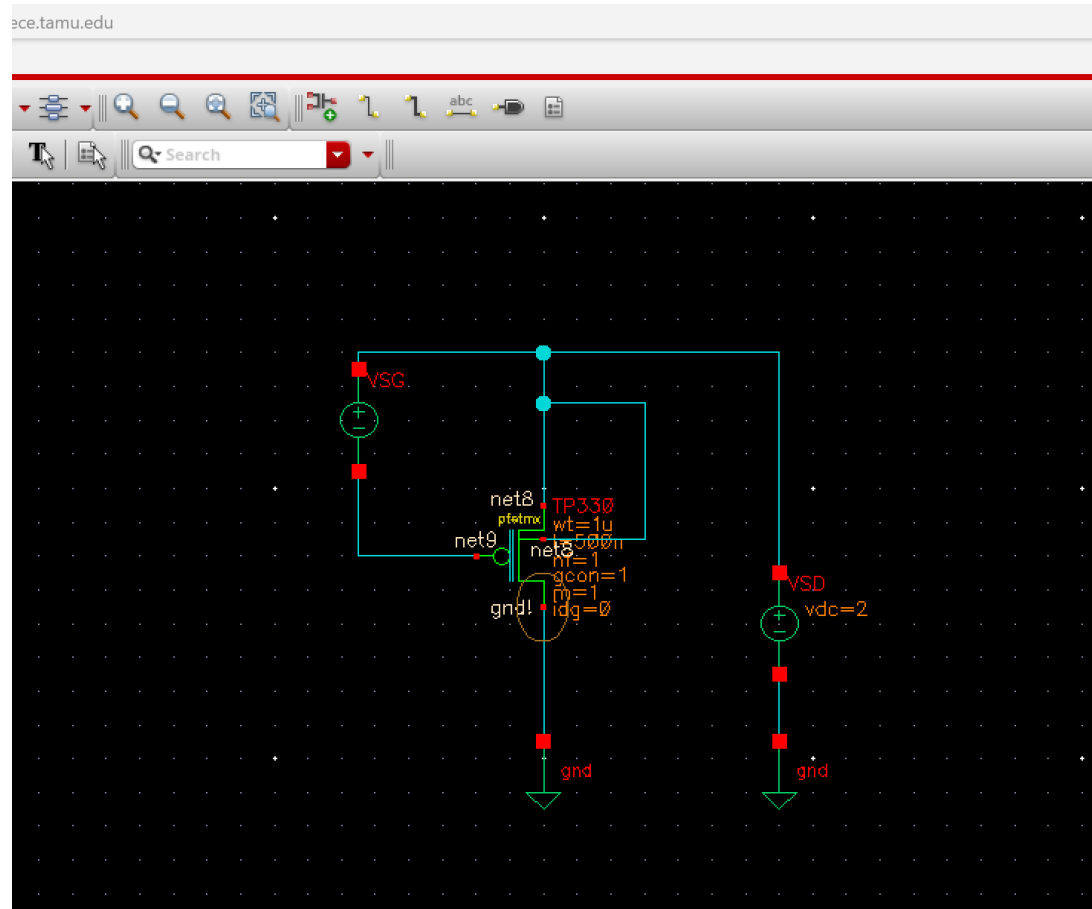
c. Find the intrinsic gain (A_i) and unity-gain frequency (f_t) using AC simulations when $V_{DS}=2V$ and $V_{GS}=V_{tn}+100mV$.



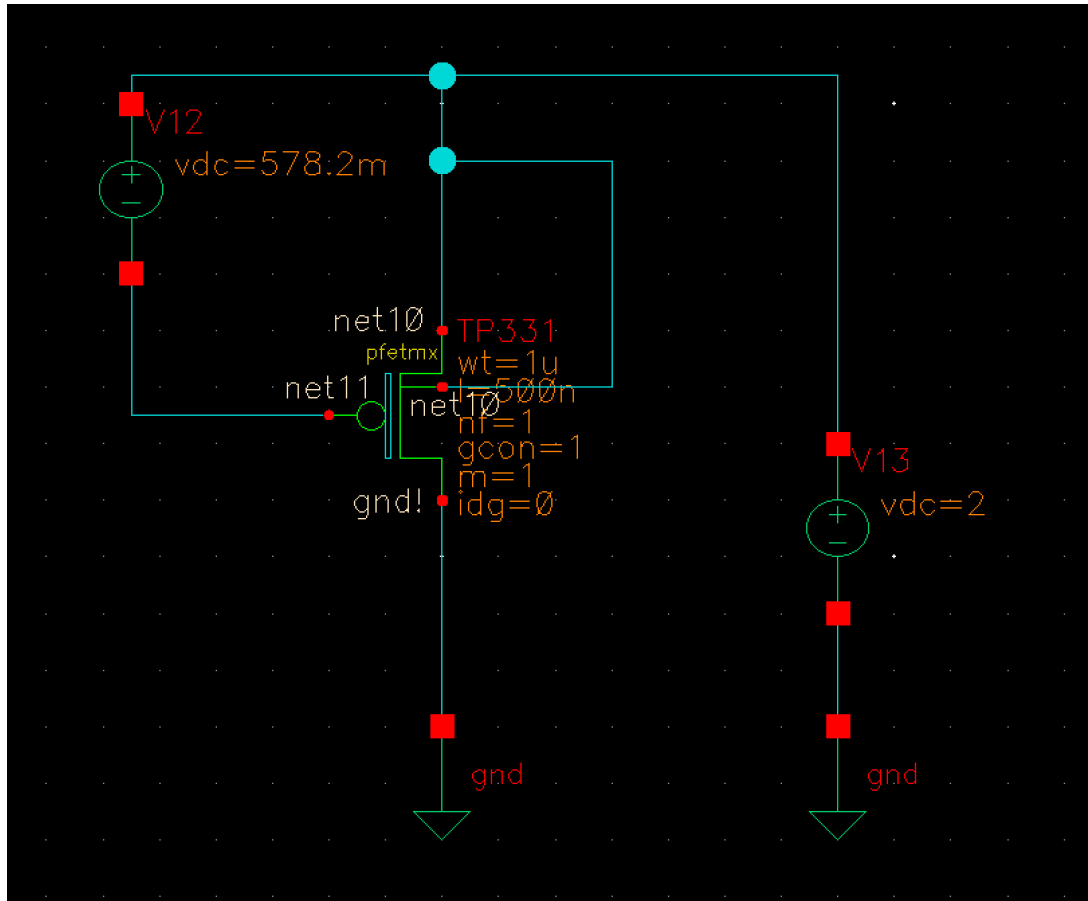
PMOS:

1. $W/L=1\mu/0.5\mu$

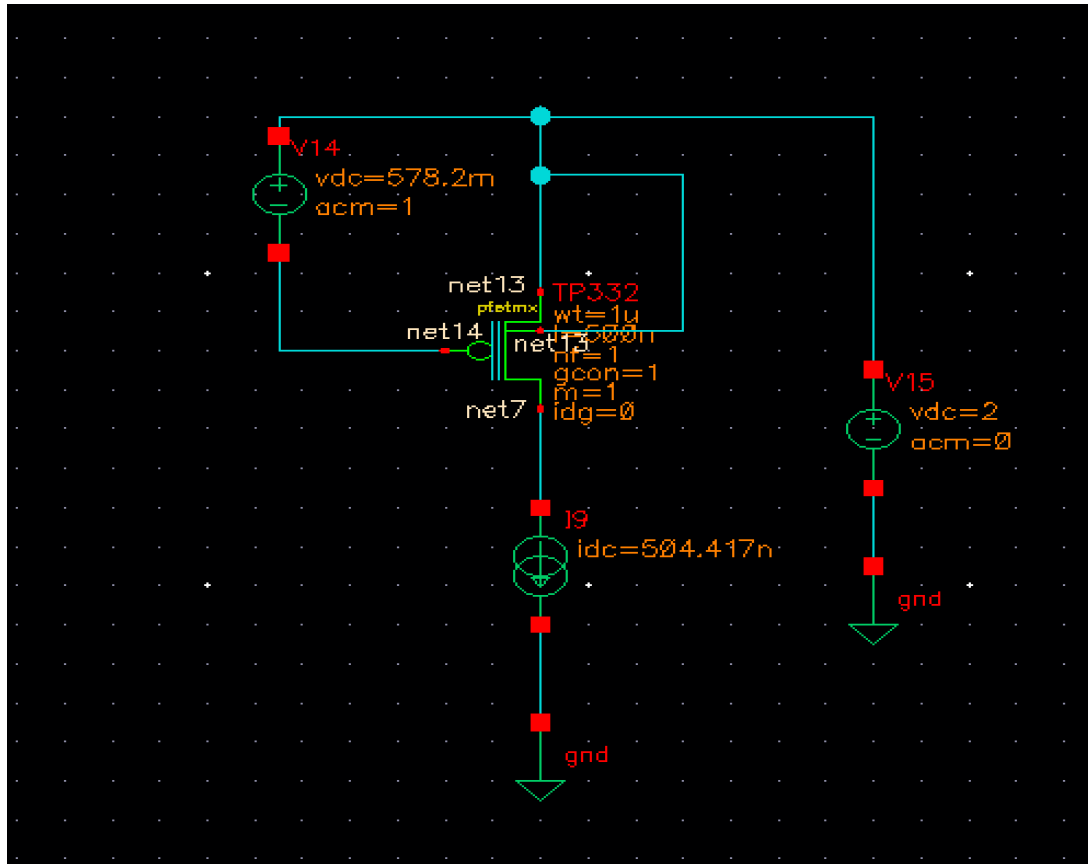
a. Fix $V_{DS}=2V$. Sweep V_{GS} from 0 to 5V, plot I_D and g_m (dI_D/dV_{GS}) vs. V_{GS} , and extract V_{tn} , $\mu_n C_{ox}$ and θ .



b. Set $V_{GS}=V_{tn}+100\text{mV}$. Sweep V_{DS} from 0 to 5V, plot and g_{ds} (dI_D/dV_{DS}) vs. V_{DS} . Plot the output resistance r_{ds} , and its first and second derivative as function of V_{DS} .

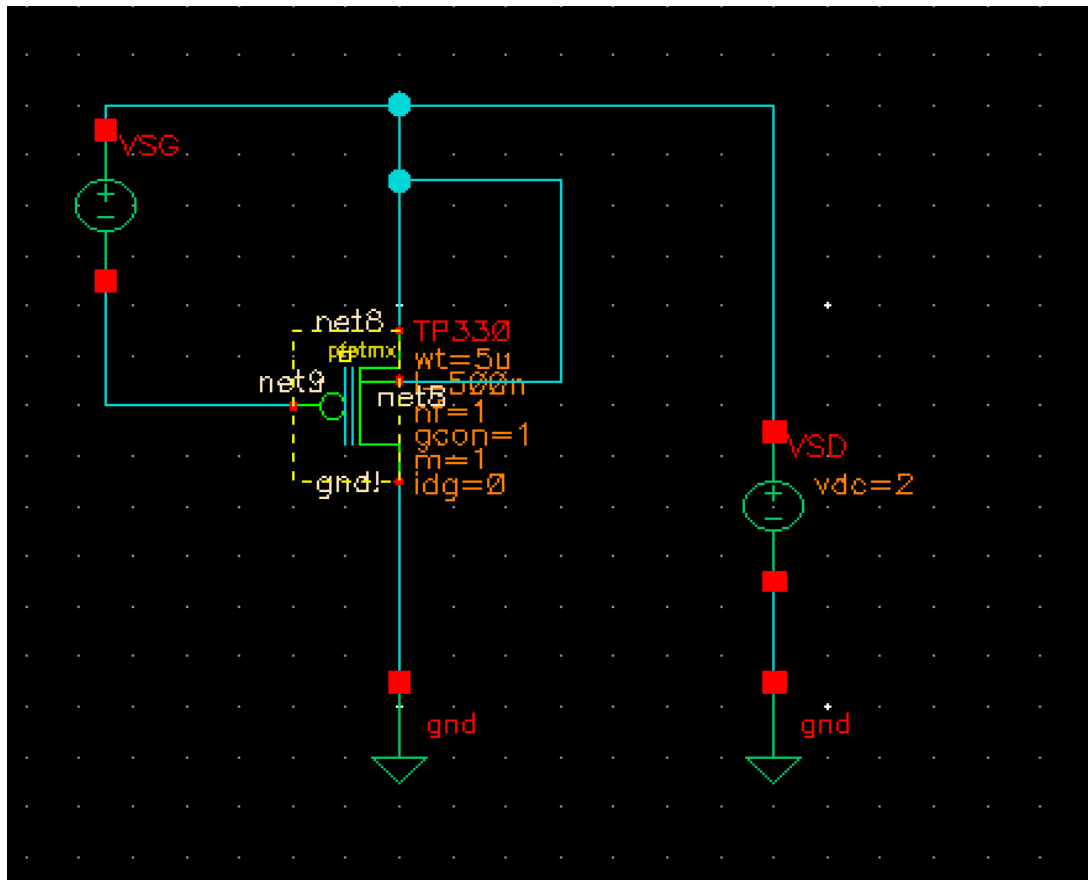


c. Find the intrinsic gain (A_i) and unity-gain frequency (f_t) using AC simulations when $V_{DS}=2V$ and $V_{GS}=V_{tn}+100mV$.

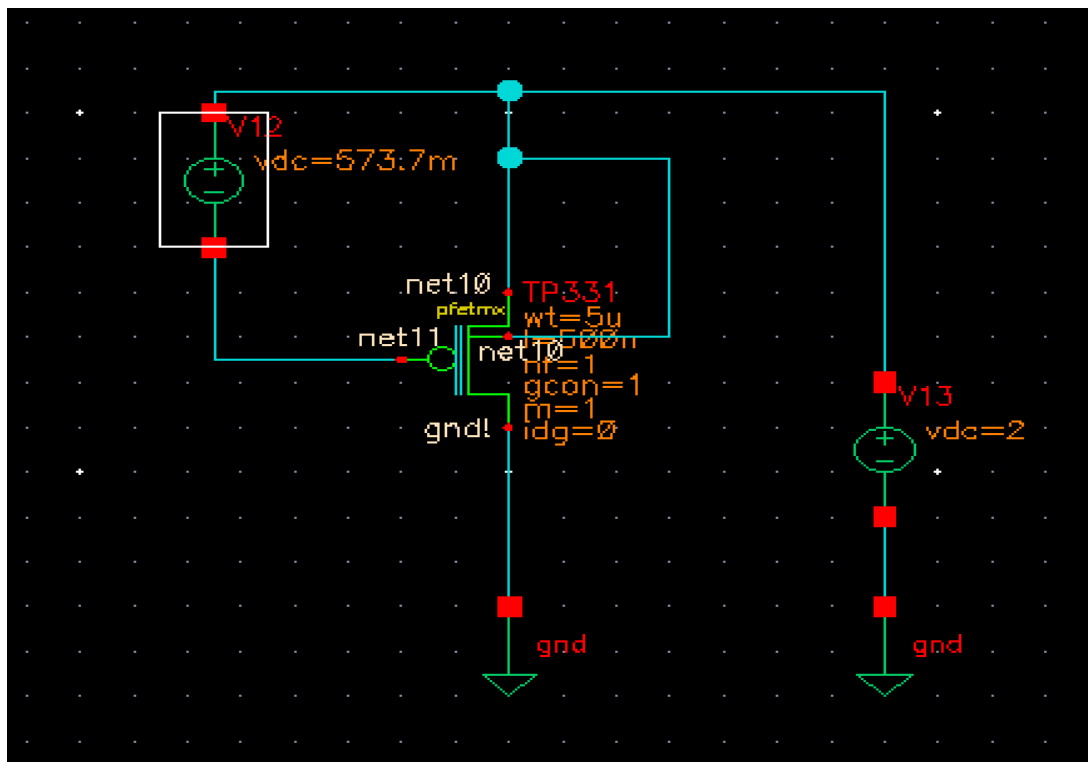


2. $W/L=5\mu/0.5\mu$

a. Fix $V_{DS}=2V$. Sweep V_{GS} from 0 to 5V, plot I_D and g_m (dI_D/dV_{GS}) vs. V_{GS} , and extract V_{tn} , $\mu_n C_{ox}$ and θ .

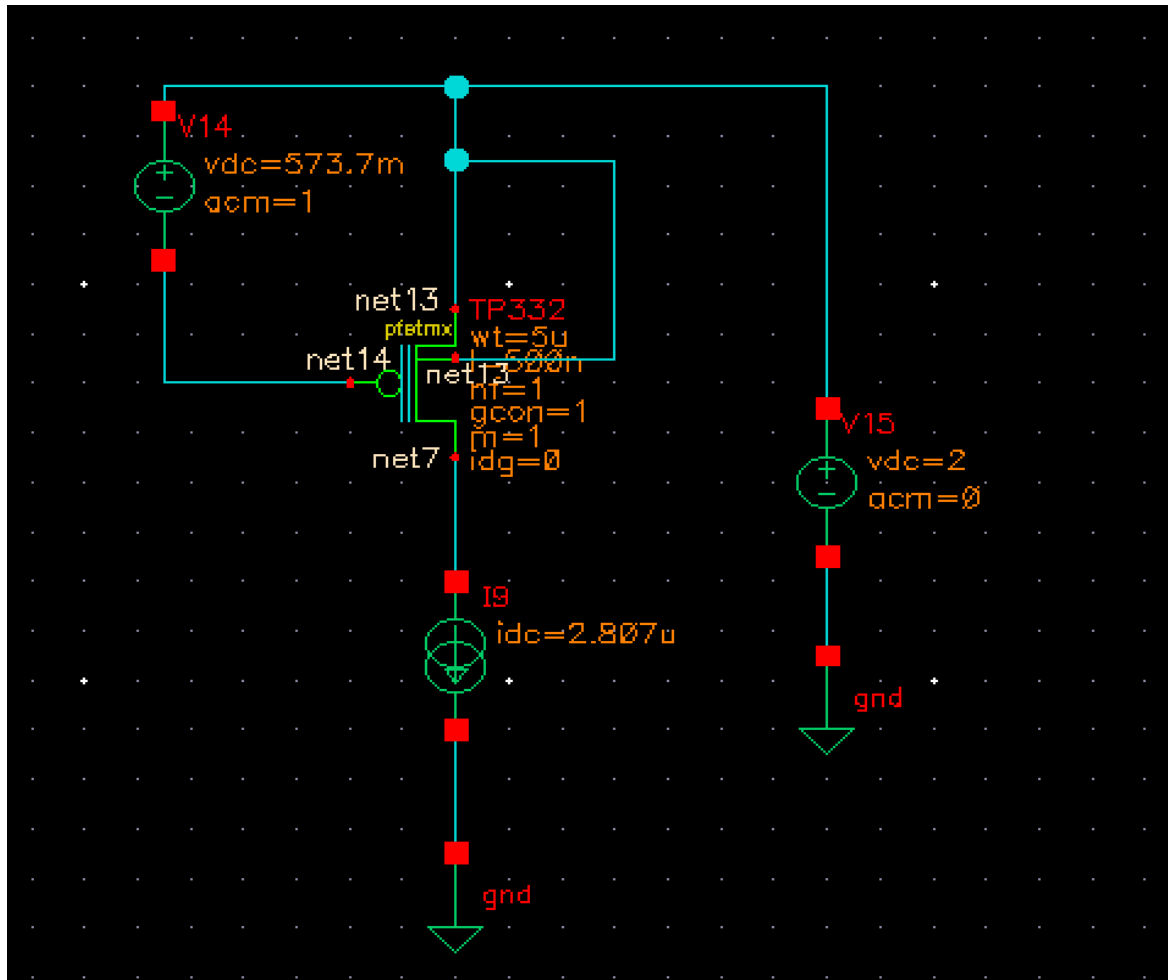


b. Set $V_{GS}=V_{tn}+100mV$. Sweep V_{DS} from 0 to 5V, plot and g_{ds} (dI_D/dV_{DS}) vs. V_{DS} . Plot the output resistance r_{ds} , and its first and second derivative as function of V_{DS} .



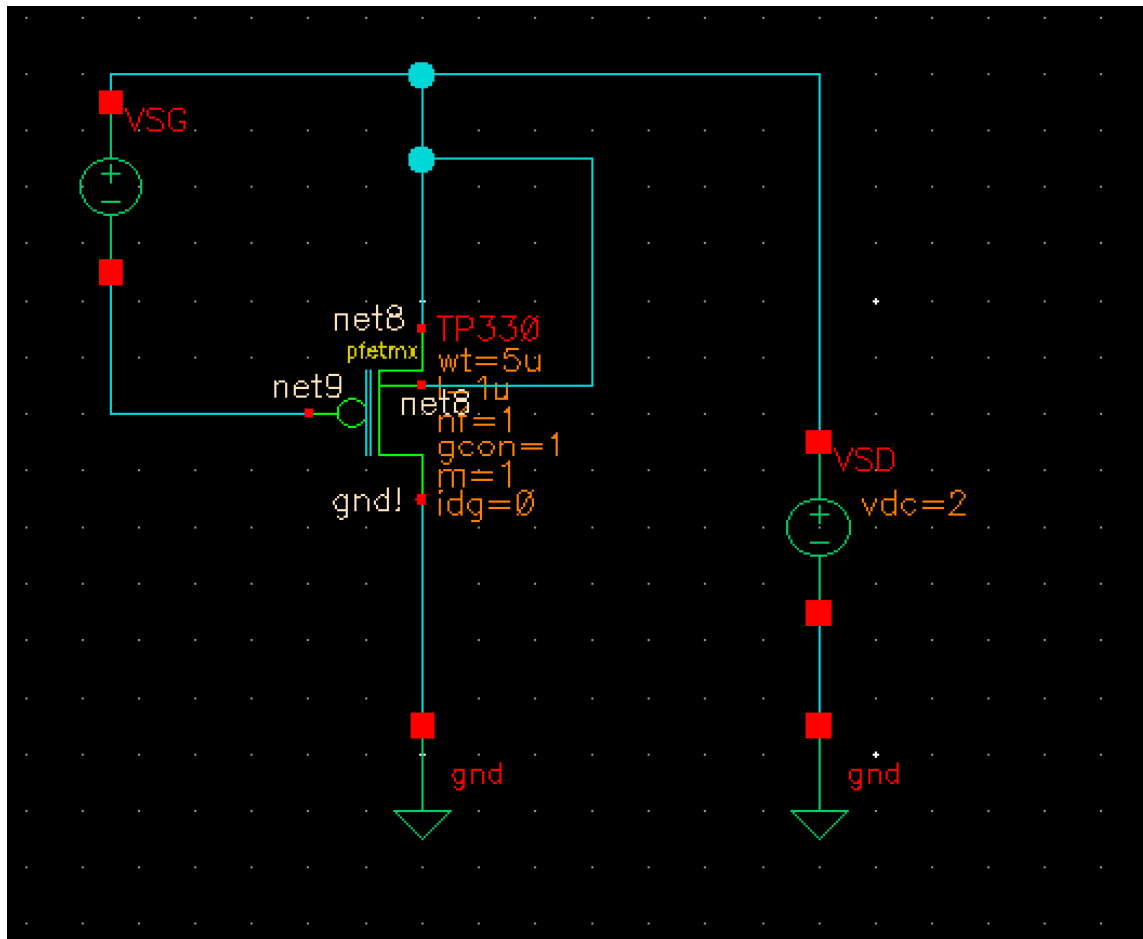
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$$V_{GS} = V_{tn} + 100\text{mV}.$$

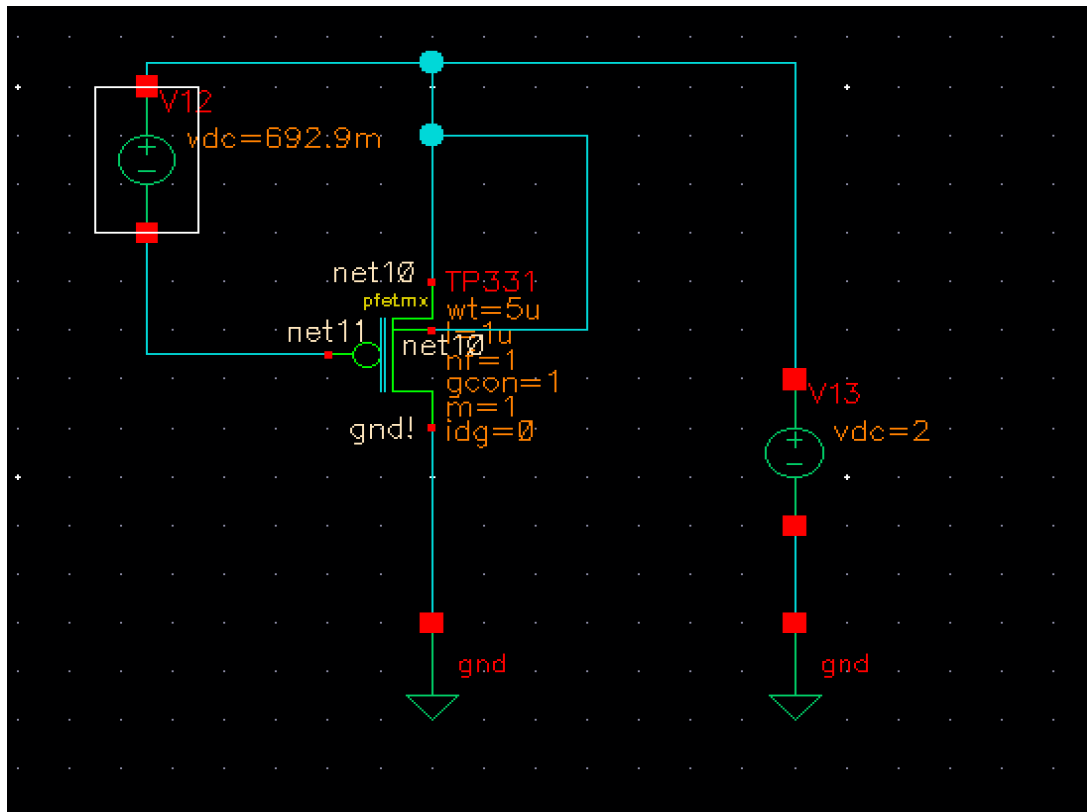


$$3. W/L = 5\mu/1\mu$$

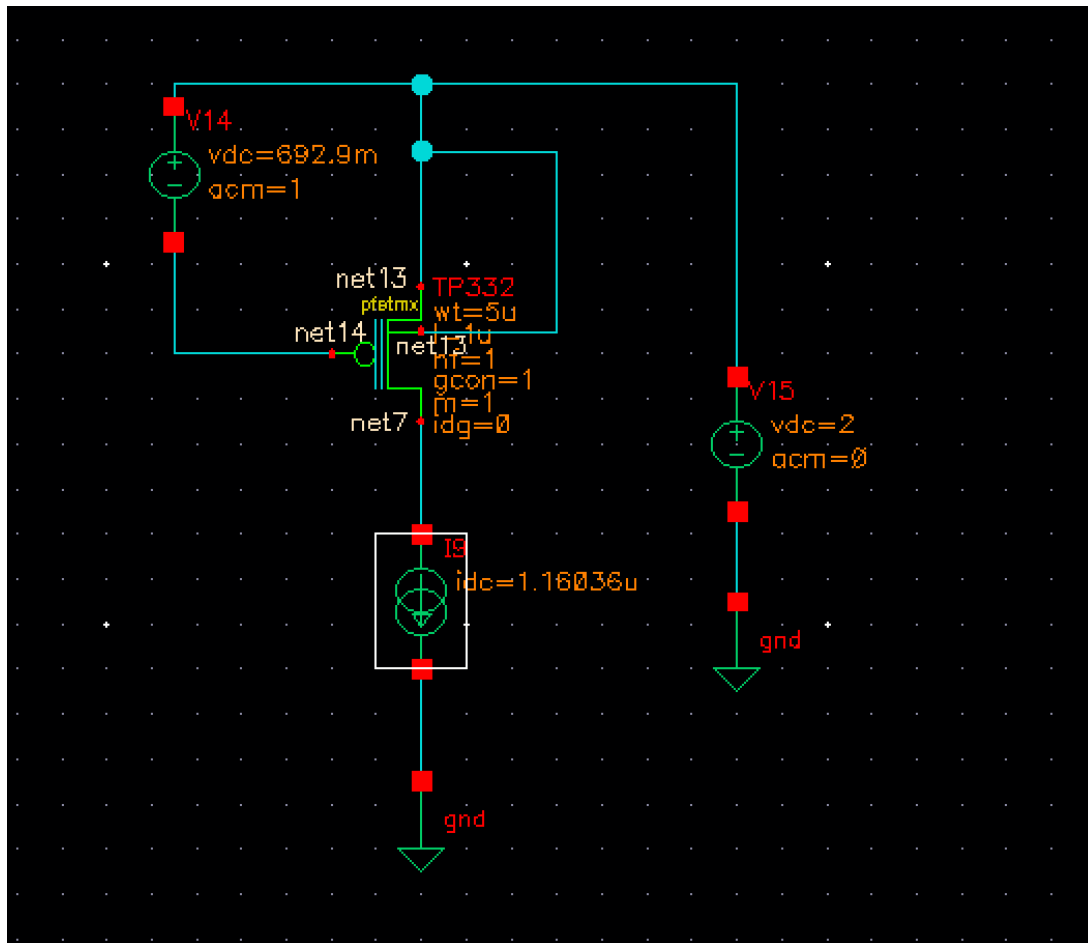
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b. Set $V_{GS}=V_{tn}+100\text{mV}$. Sweep V_{DS} from 0 to 5V, plot and gds (dI_D/dV_{DS}) vs. V_{DS} . Plot the output resistance r_{ds} , and its first and second derivative as function of V_{DS} .



c. Find the intrinsic gain (A_i) and unity-gain frequency (f_t) using AC simulations when $V_{DS}=2V$ and $V_{GS}=V_{tn}+100mV$.

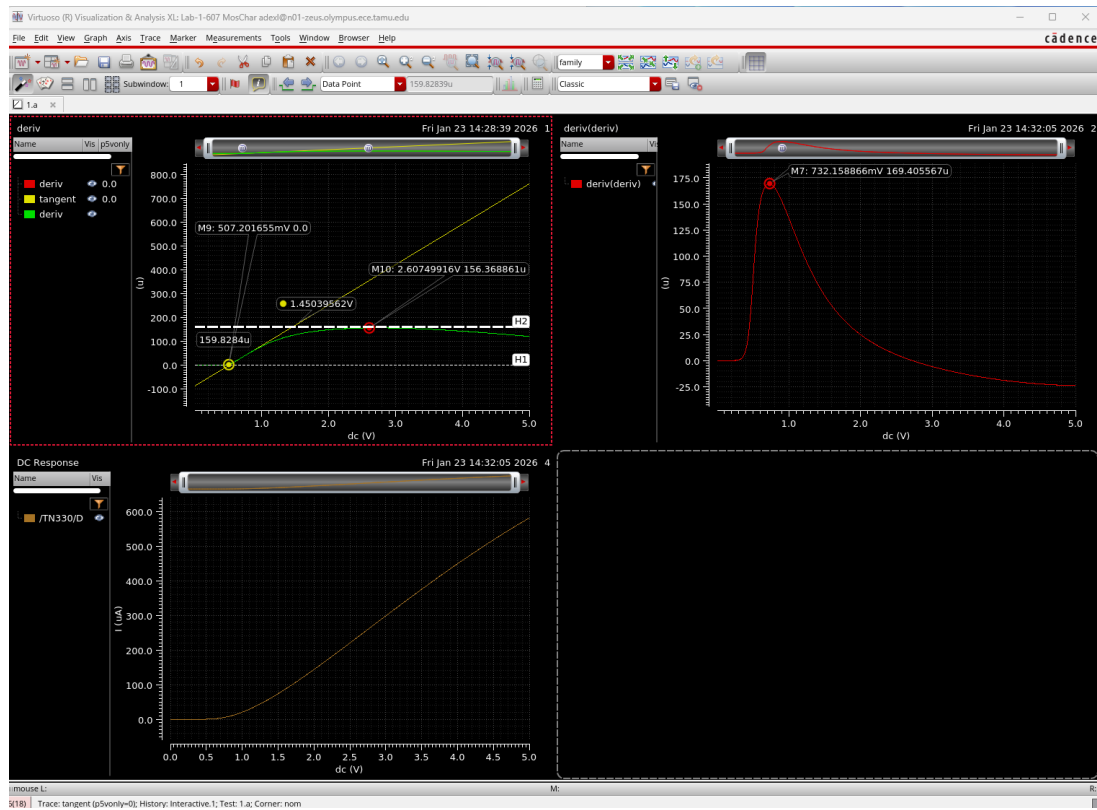


VI. SIMULATION RESULTS

Images of simulation results requested in the lab manual should be included here. Plots should be annotated to show relevant data points which will include either (1) data points that are explicitly requested in the lab manual or (2) data points from which other requested parameters are calculated. Data in the images should also be legible, so scale the images accordingly to make sure the data is easy to read. Post-layout simulations should be plotted in the same window with the pre-layout simulation results in order to make the differences between the two more apparent.

1. $W/L = 1\mu/0.7\mu$

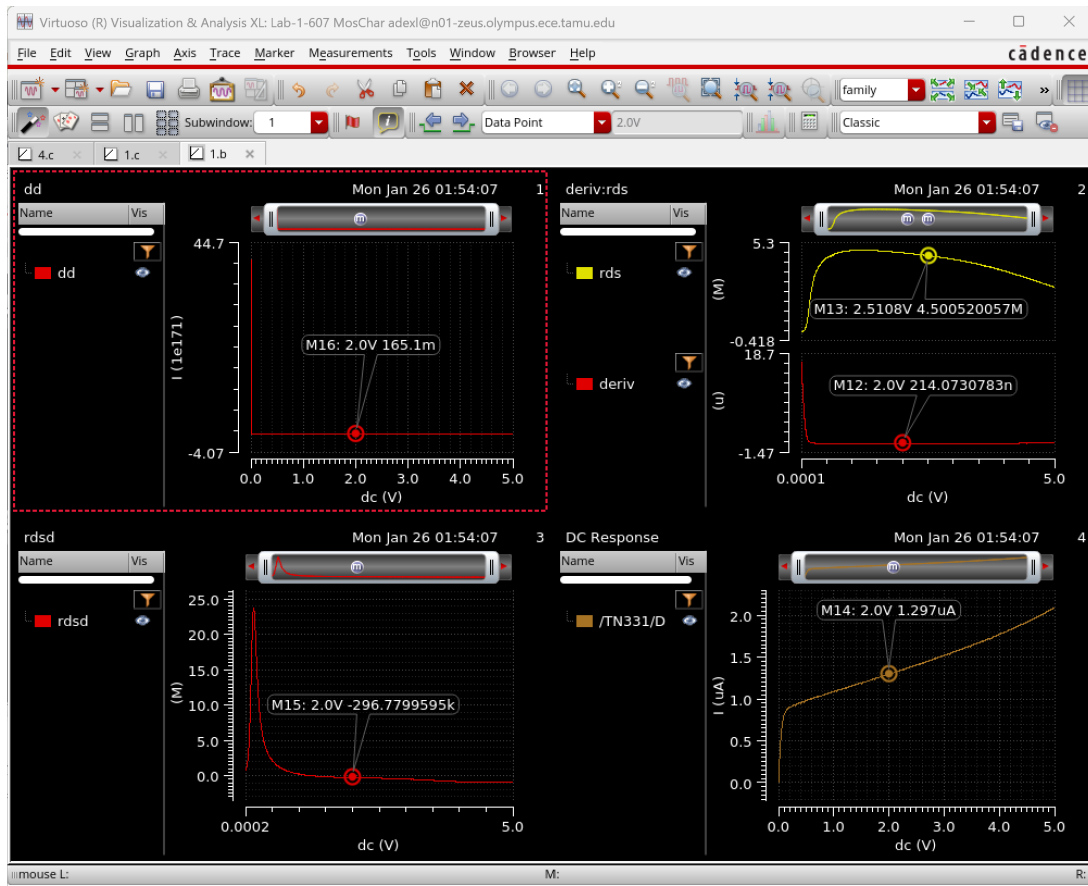
a. Fix $V_{DS} = 2V$. Sweep V_{GS} from 0 to 5V, plot I_D and g_m (dI_D/dV_{GS}) vs. V_{GS} , and extract V_{tn} , $\mu_n C_{ox}$ and θ .



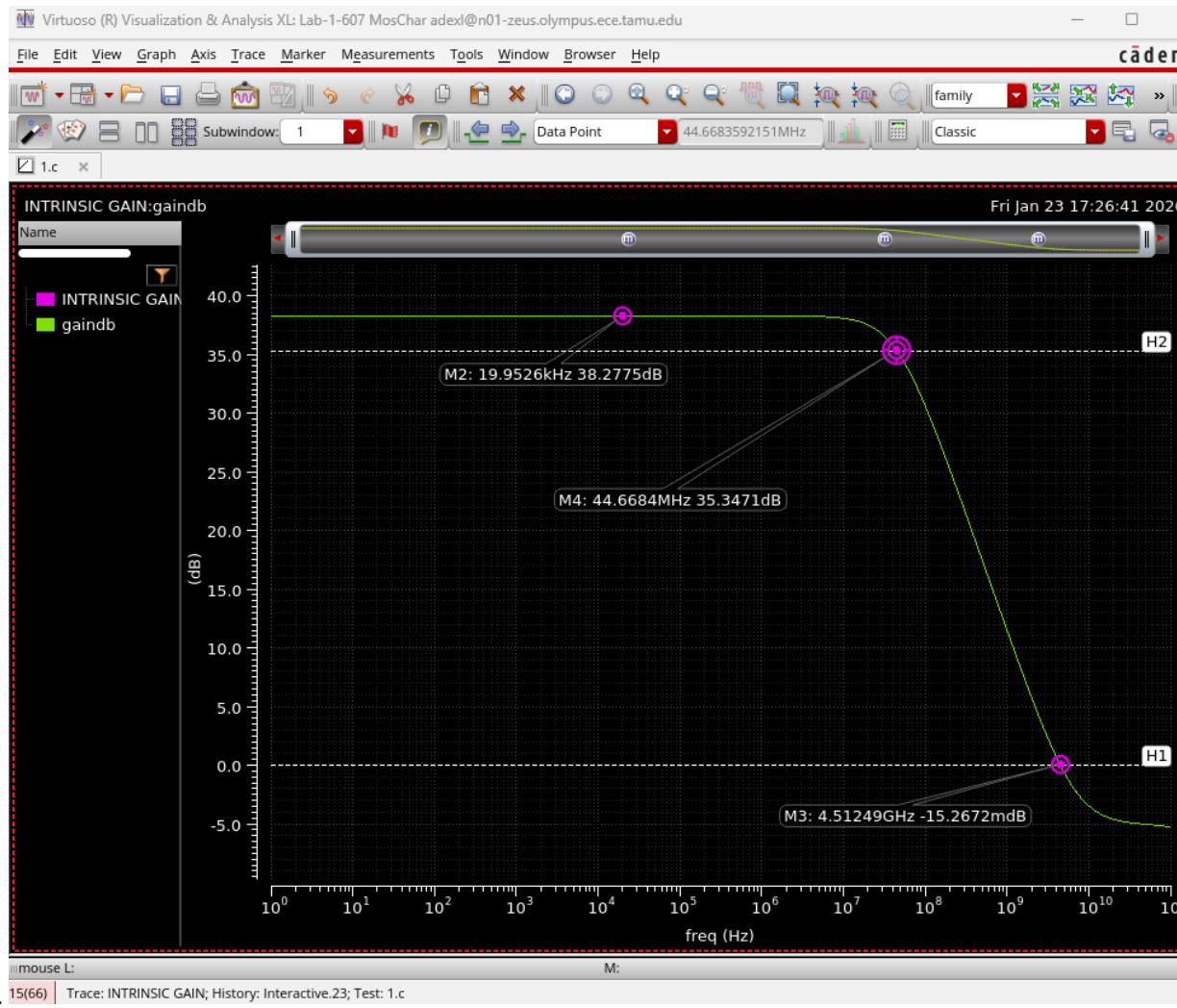
Test	Output	Nominal	Spec	Weight	Pass/Fail
1.a	ID				
1.a	deriv				
1.a	deriv(deriv)				
1.a	y _{max}	169.5u			
1.a	x _{value}	724m			
1.a	y _{value}	36.74u			
1.a	tangent				
1.a	v _t	507.2m			
1.a	y _{max2}	156.5u			
1.a	v _t +(1/2theta)	1.431			

b. Set $V_{GS}=V_{tn}+100\text{mV}$. Sweep V_{DS} from 0 to 5V, plot and gds (dI_D/dV_{DS}) vs. V_{DS} . Plot the output resistance r_{ds} , and its first and second derivative as function of V_{DS} .

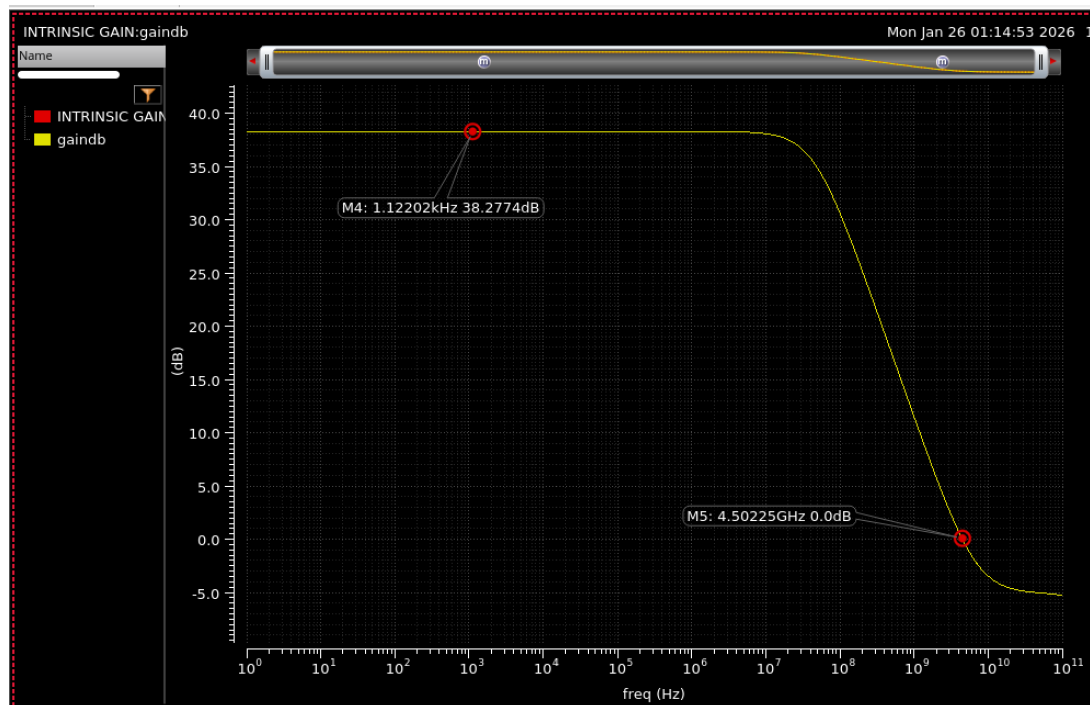
Including r_{ds} and its derivative



c. Find the intrinsic gain (A_i) and unity-gain frequency (f_t) using AC simulations when $V_{DS}=2V$ and

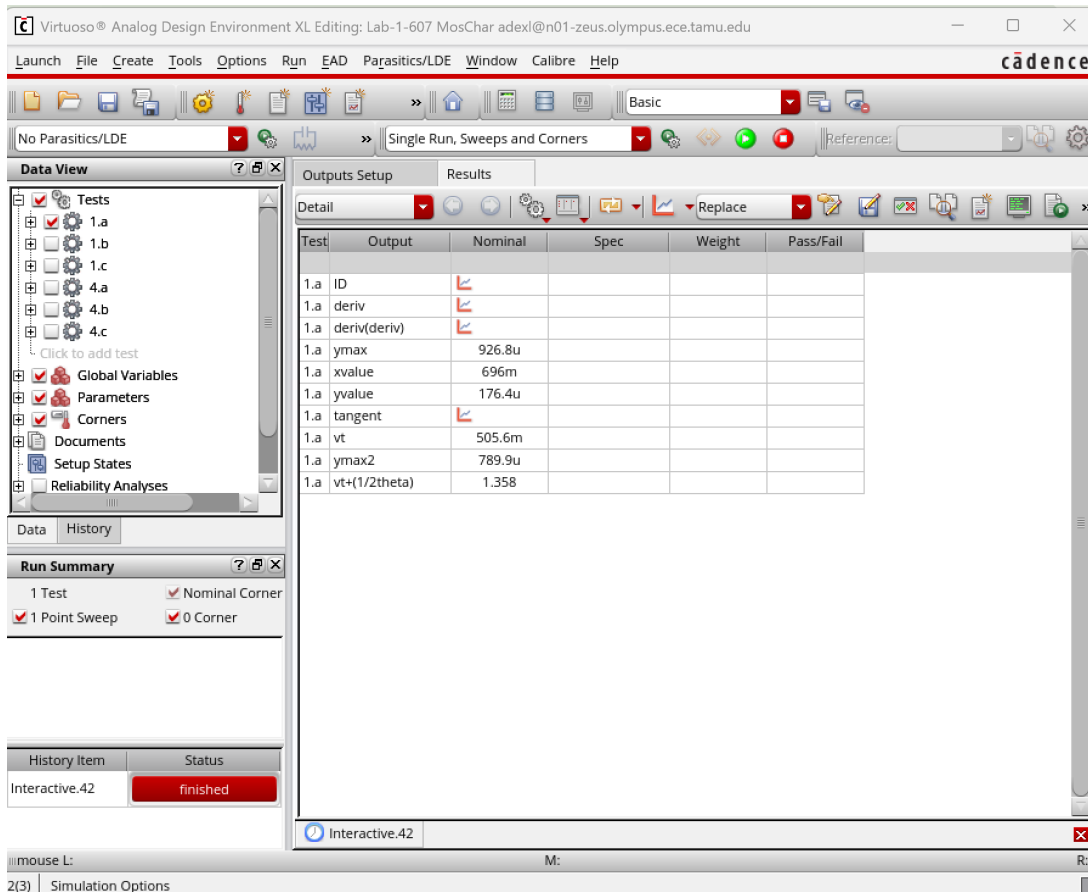
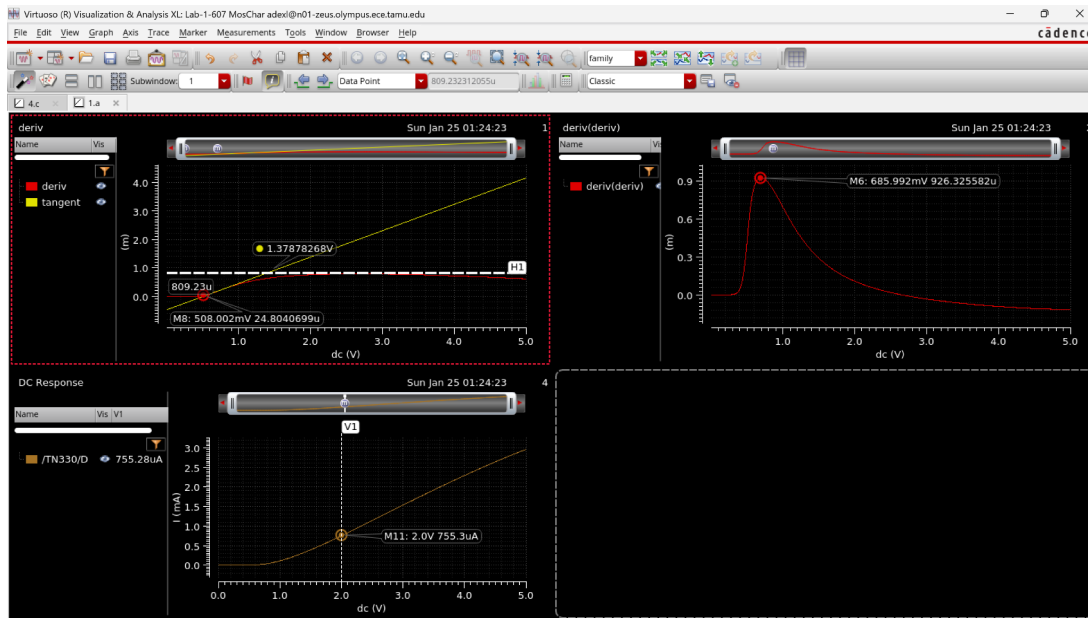


$V_{GS} = V_{tn} + 100\text{mV}$.

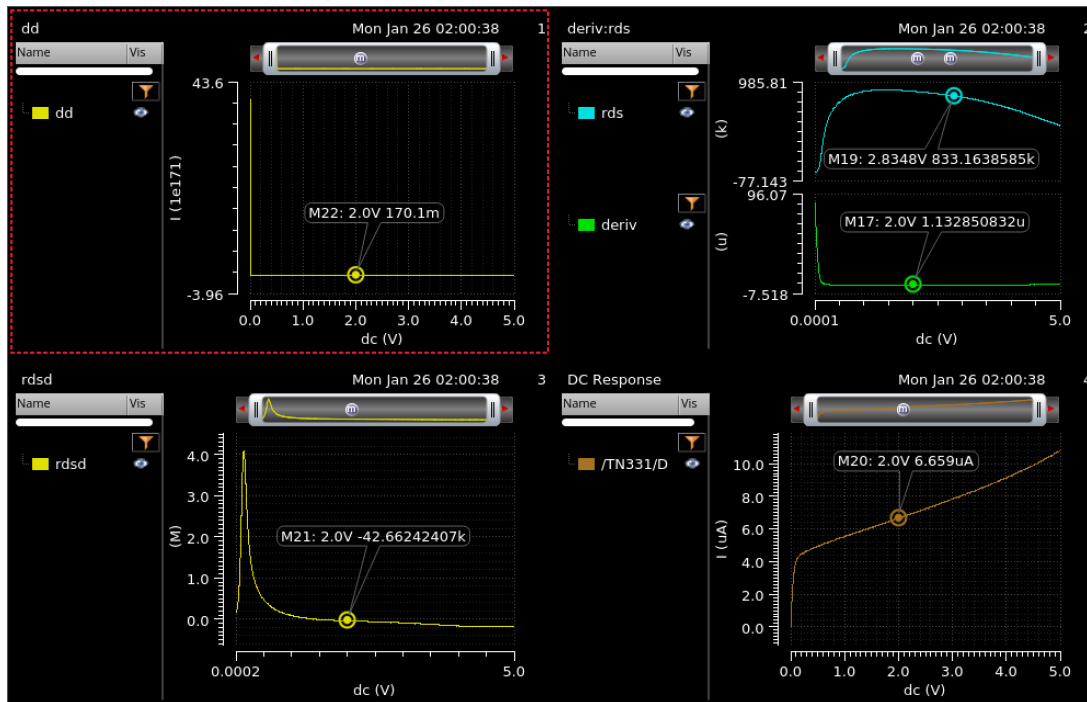


$$2.W/L=5\mu/0.7\mu.2$$

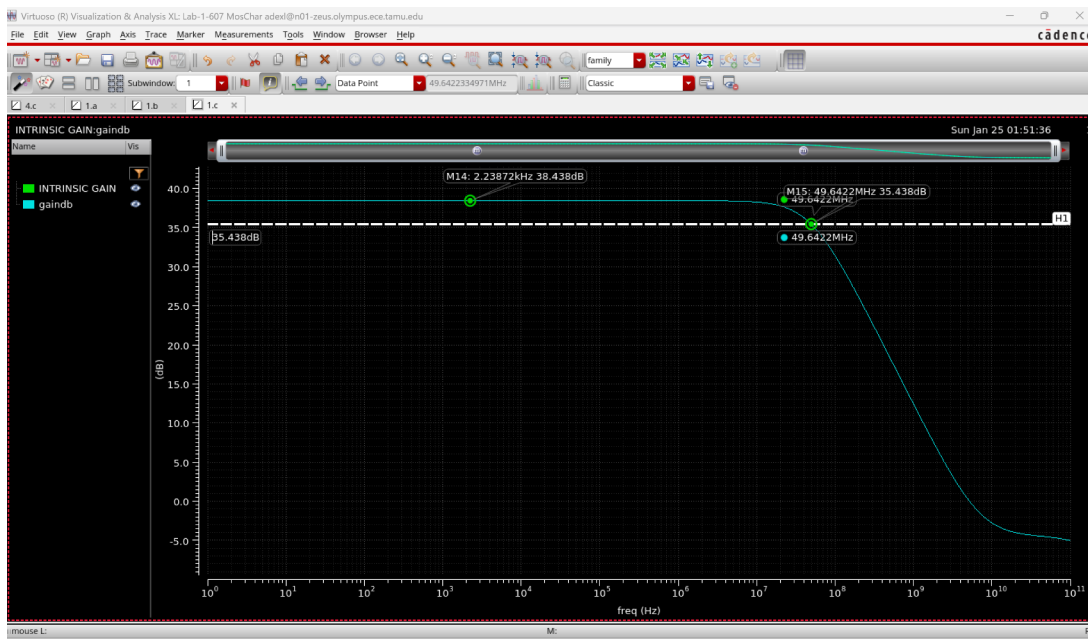
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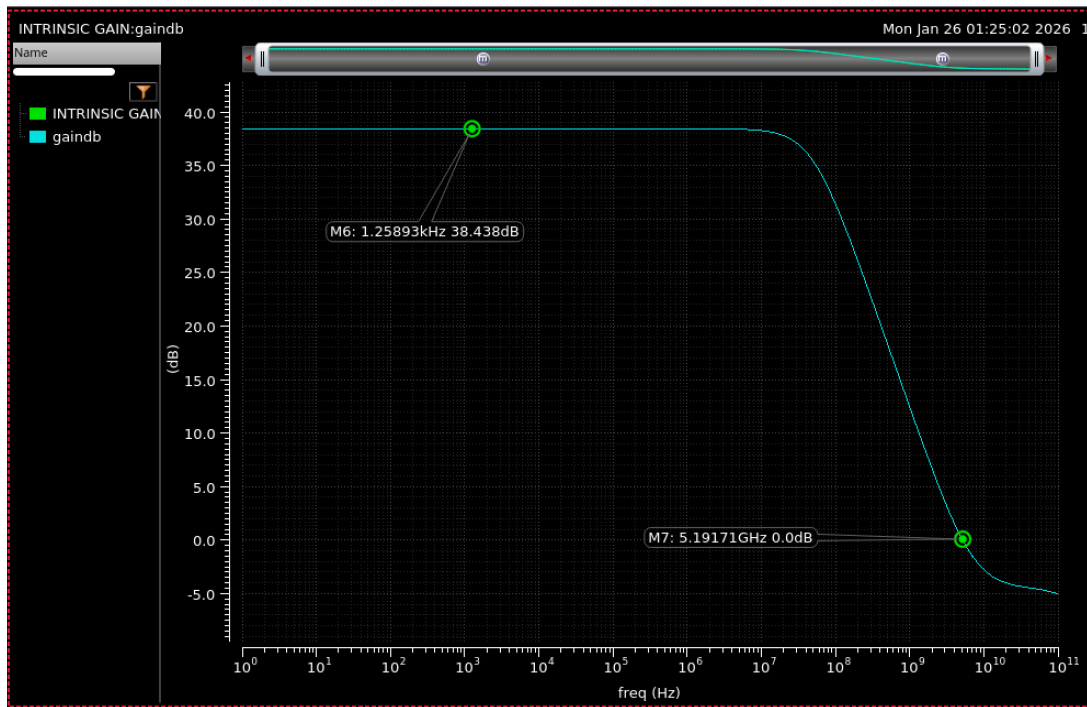


b. Set $V_{GS}=V_{tn}+100mV$. Sweep V_{DS} from 0 to 5V, plot and g_{ds} (dI_D/dV_{DS}) vs. V_{DS} . Plot the output resistance r_{ds} , and its first and second derivative as function of V_{DS} .



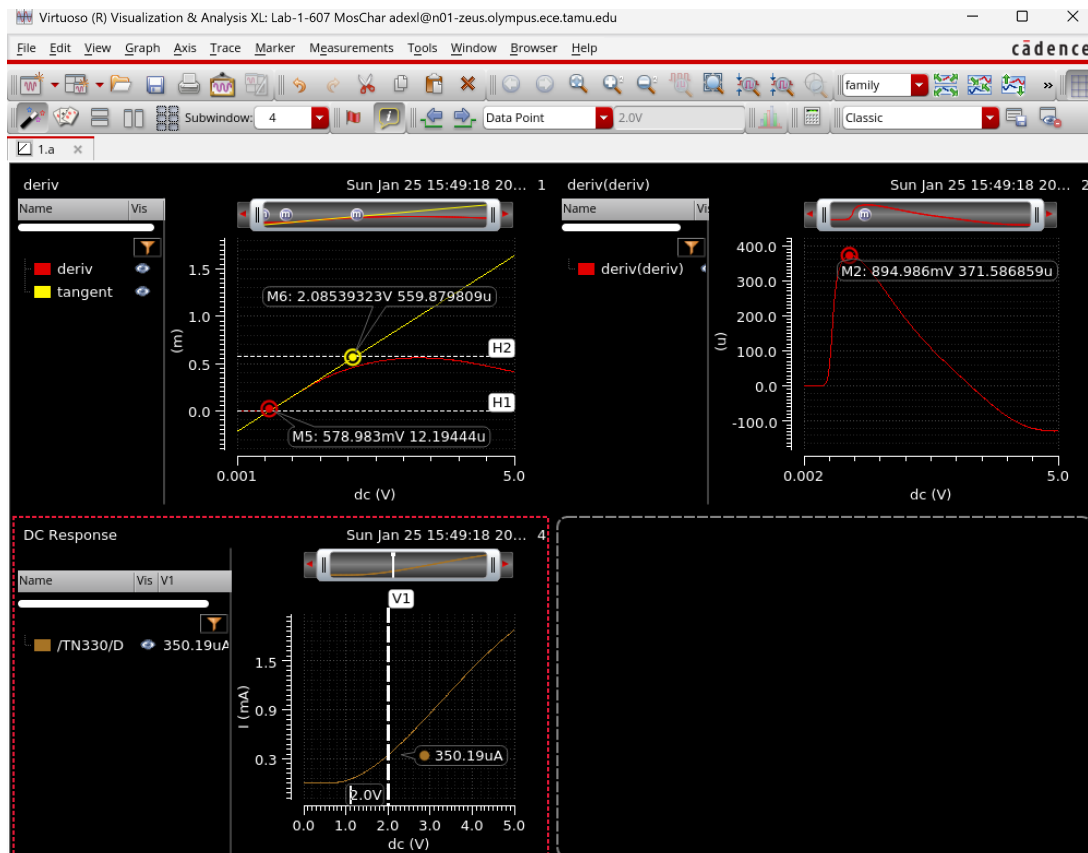
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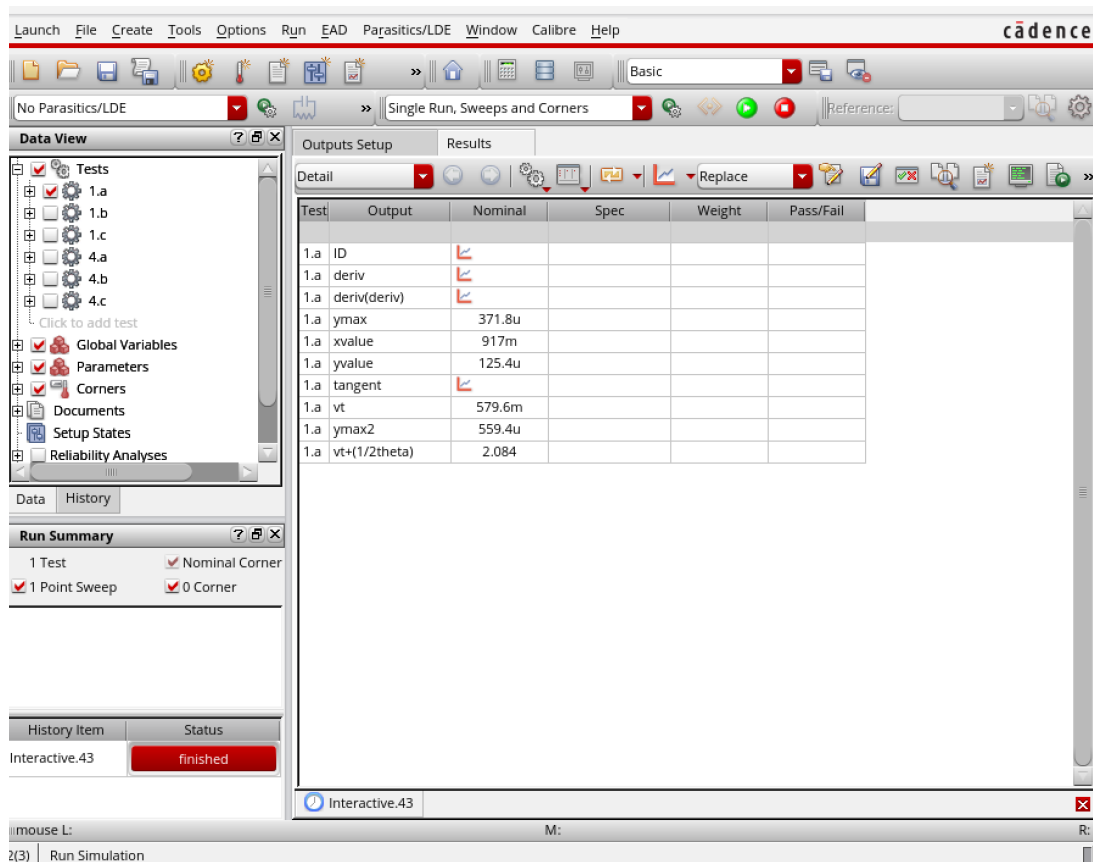




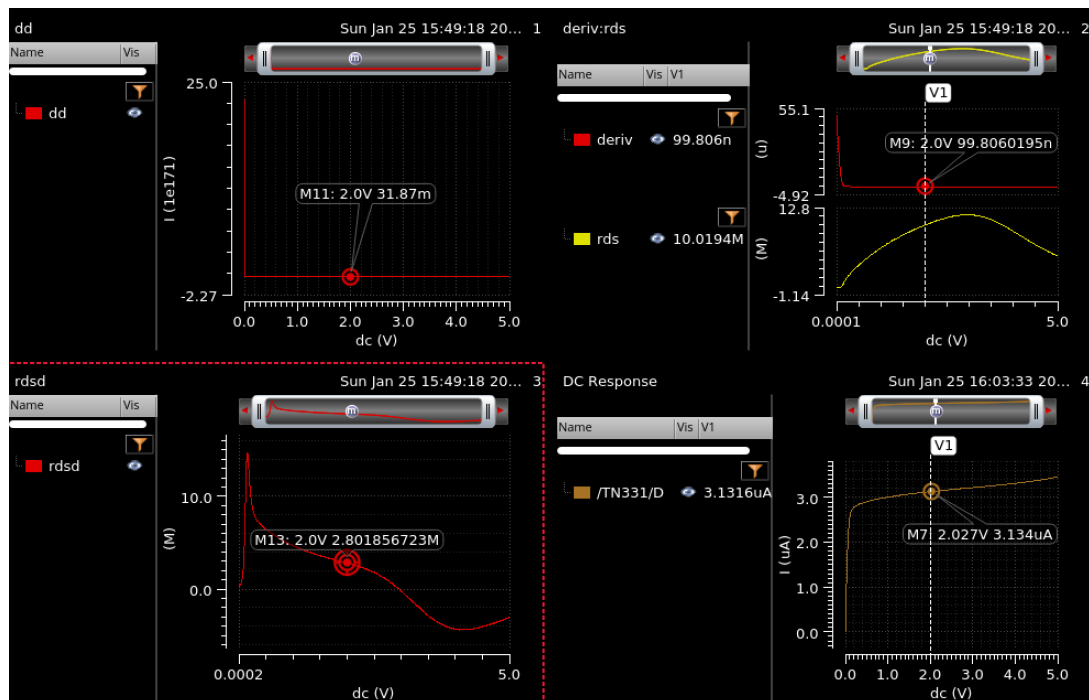
3. $W/L = 5\mu/1.4\mu$.

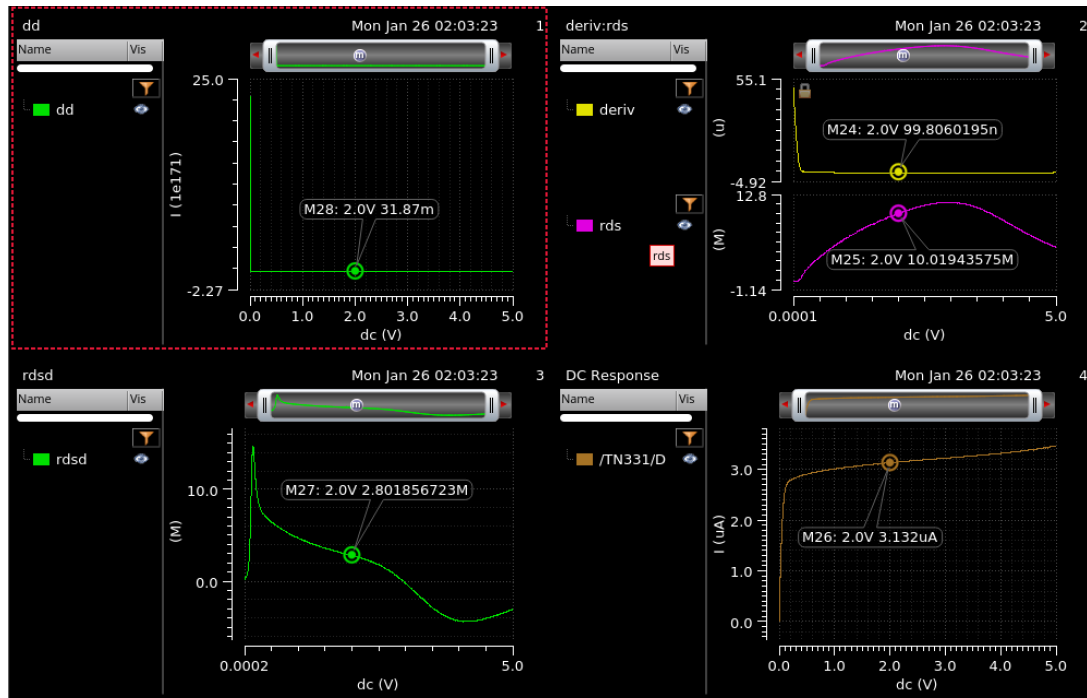
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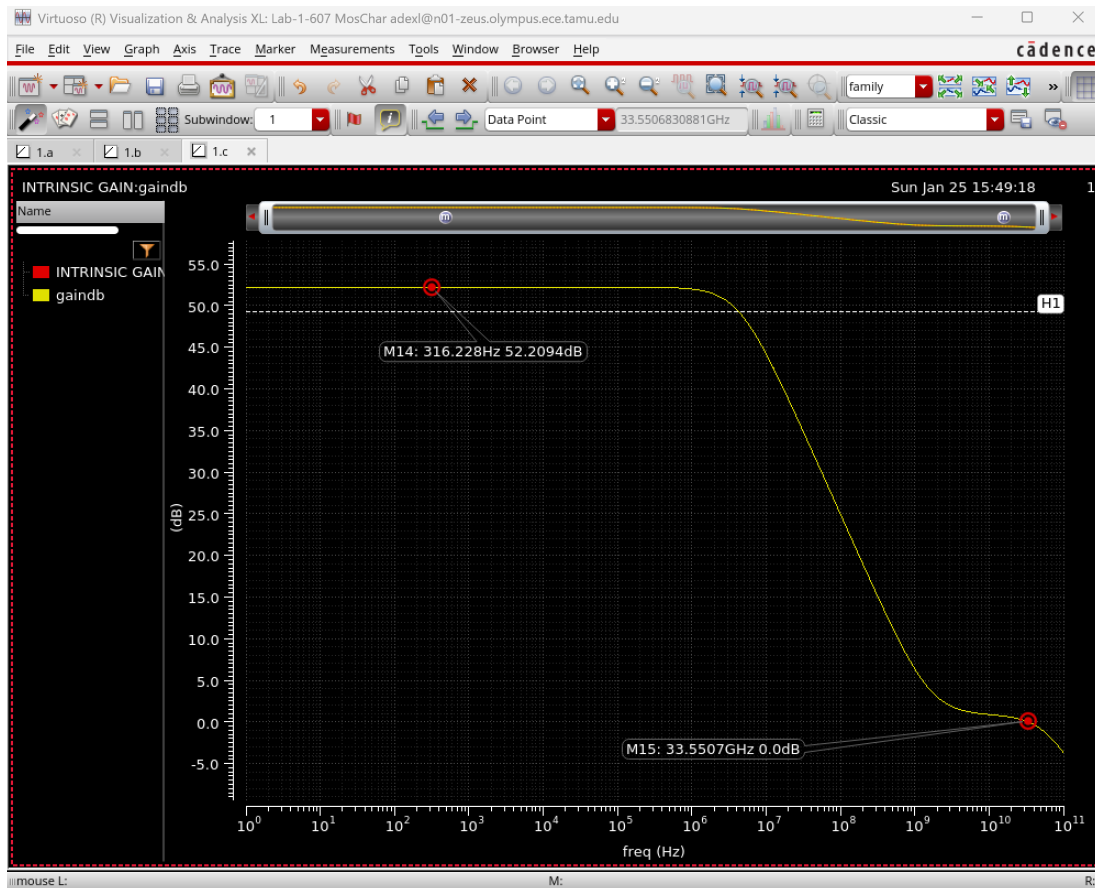


b. Set $V_{GS} = V_{tn} + 100\text{mV}$. Sweep V_{DS} from 0 to 5V, plot and g_{ds} (dI_D/dV_{DS}) vs. V_{DS} . Plot the output resistance r_{ds} , and its first and second derivative as function of V_{DS} .





c. Find the intrinsic gain (A_i) and unity-gain frequency (f_t) using AC simulations when $V_{DS}=2V$ and $V_{GS}=V_{tn}+100mV$.



PMOS:

$$1.W/L=1\mu/0.5\mu$$

a. Fix $V_{DS}=2V$. Sweep V_{GS} from 0 to 5V, plot I_D and g_m (dI_D/dV_{GS}) vs. V_{GS} , and extract V_{tn} , $\mu_n C_{ox}$ and θ .



Single Run, Sweeps and Corners

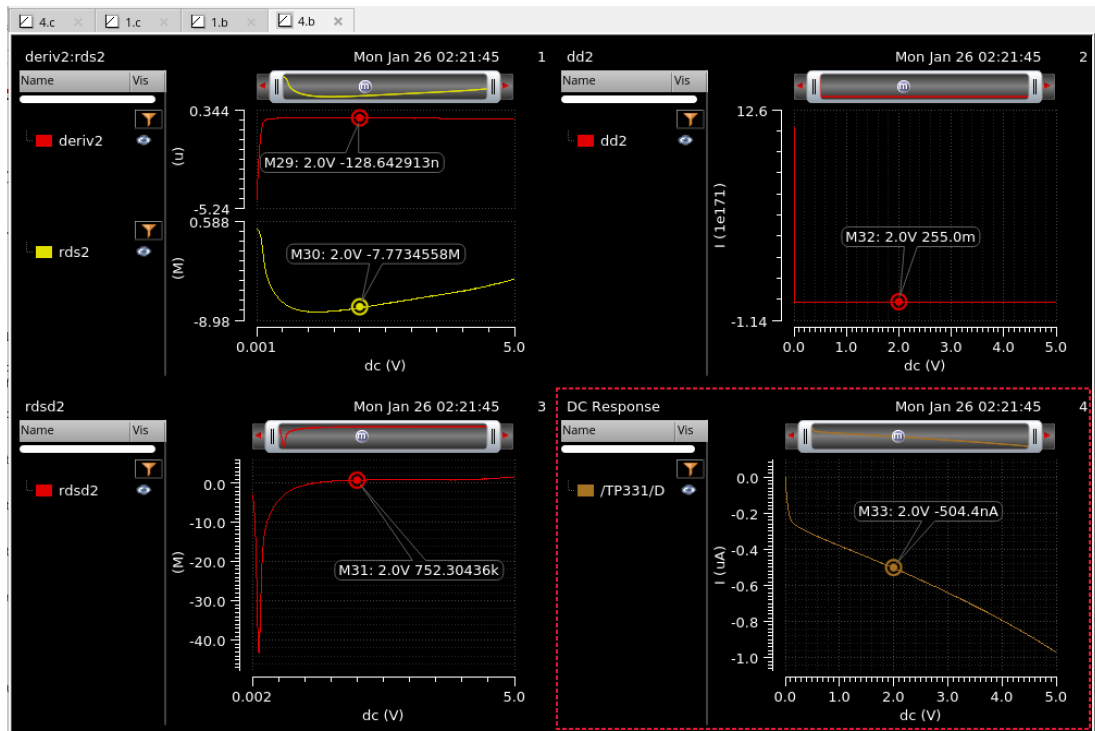
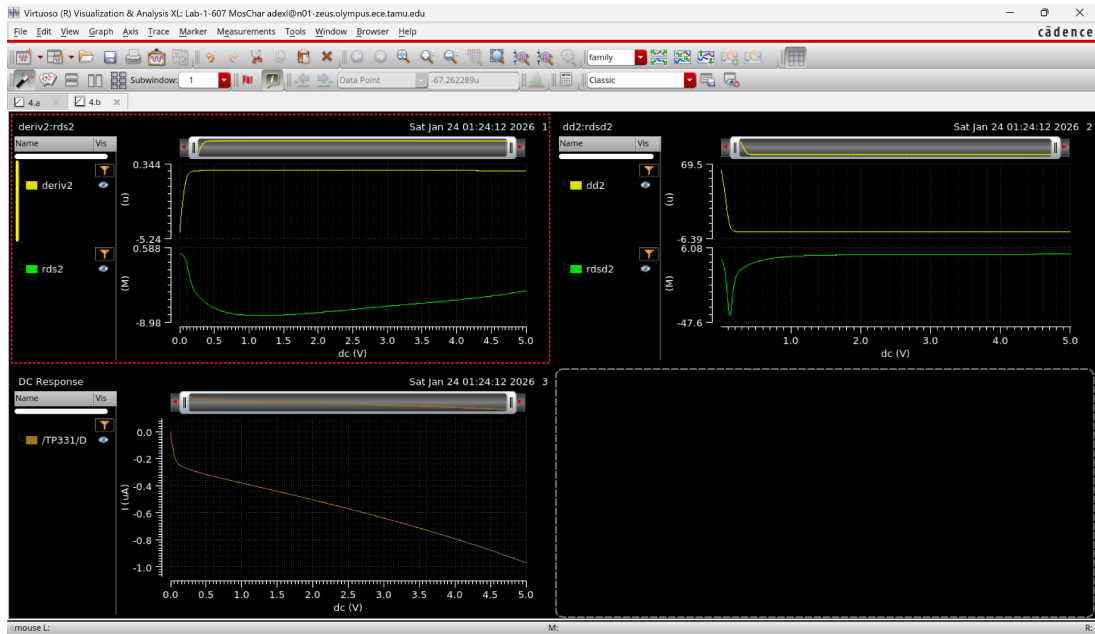
adexl x MosChar x MosChar x

Outputs Setup Results

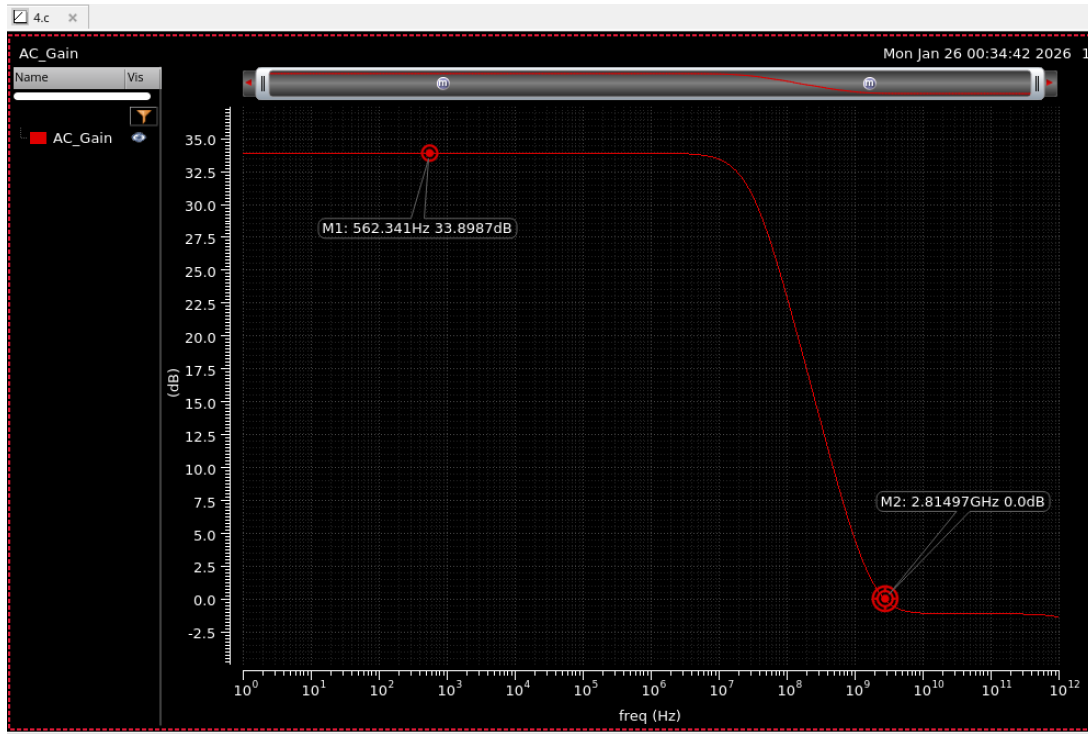
Detail

Test	Output	Nominal	Spec	Weight	Pass/Fail
4.a	ID(PMOS)				
4.a	deriv				
4.a	dd				
4.a	ymin	-60.63u			
4.a	xvalue	708m			
4.a	yvalue	-13.93u			
4.a	tangent				
4.a	ymin(d)				
4.a	vtp+(1/2theta)	1.478			
4.a	vtp	478.2m			

b. Set $V_{GS}=V_{tn}+100mV$. Sweep V_{DS} from 0 to 5V, plot g_{ds} (dI_D/dV_{DS}) vs. V_{DS} . Plot the output resistance r_{ds} , and its first and second derivative as function of V_{DS} .

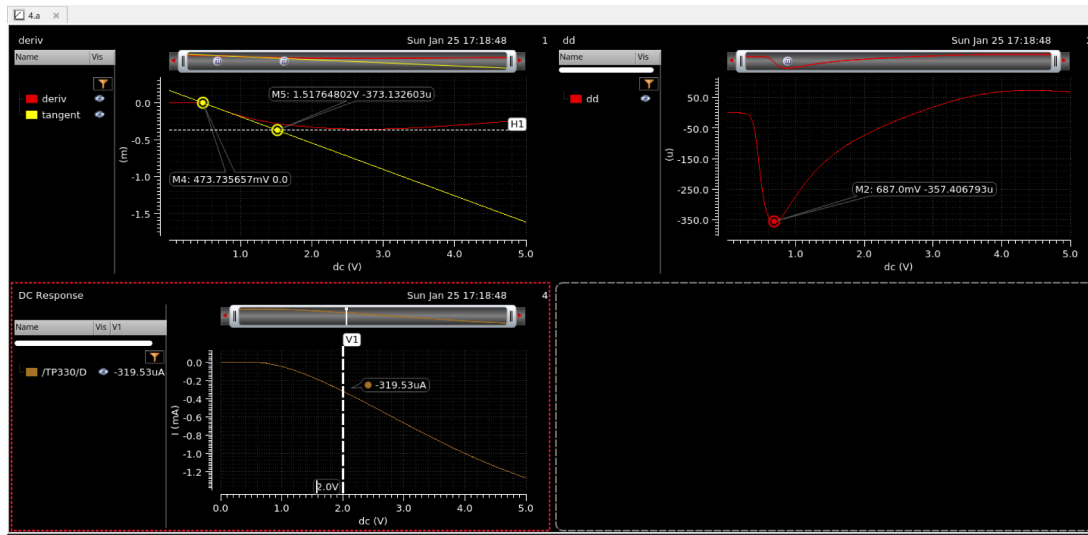


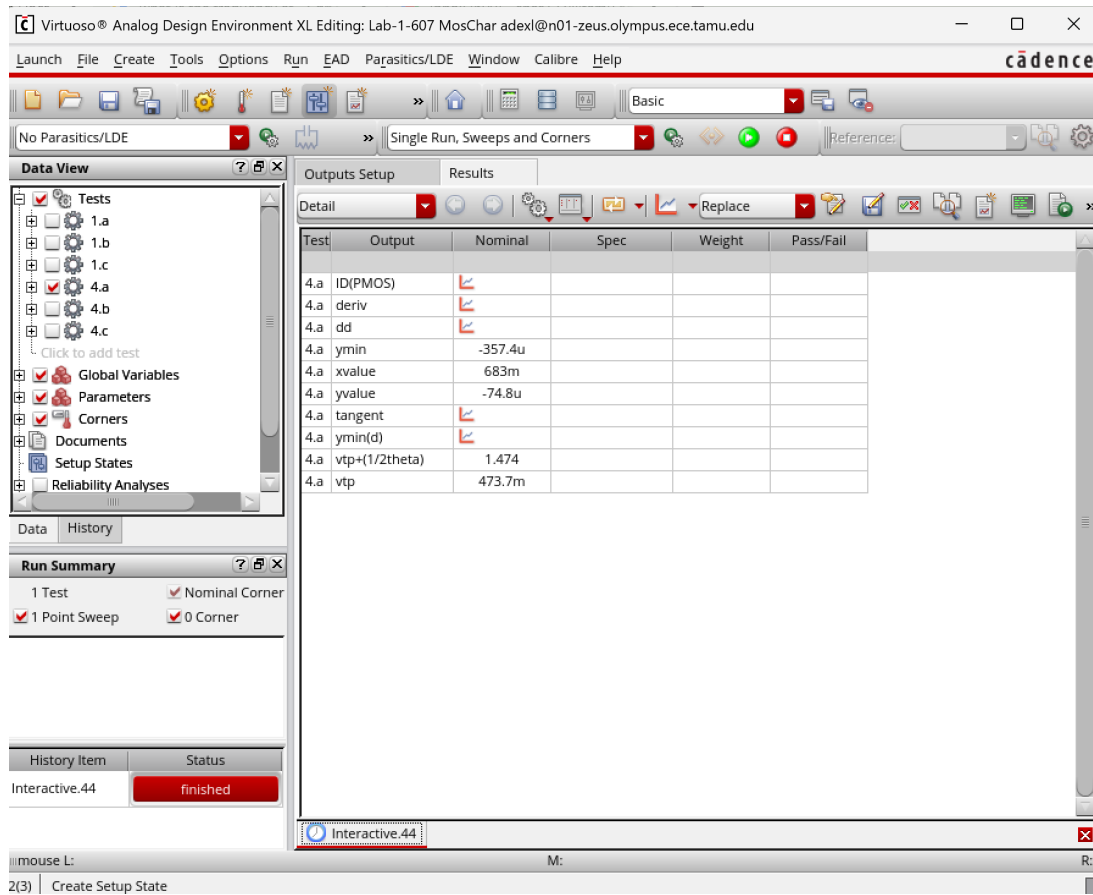
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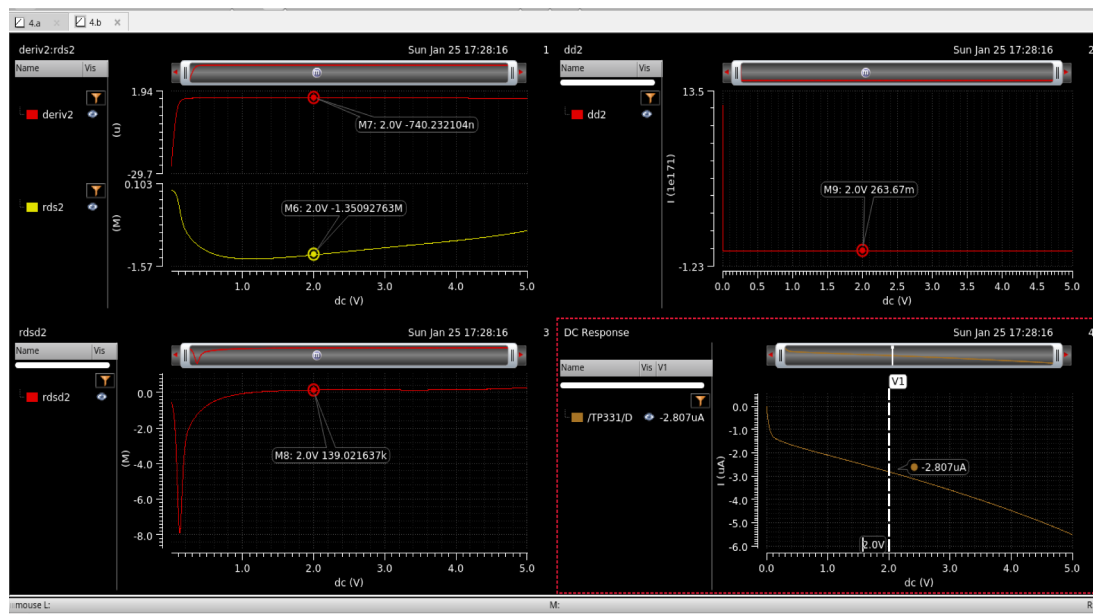
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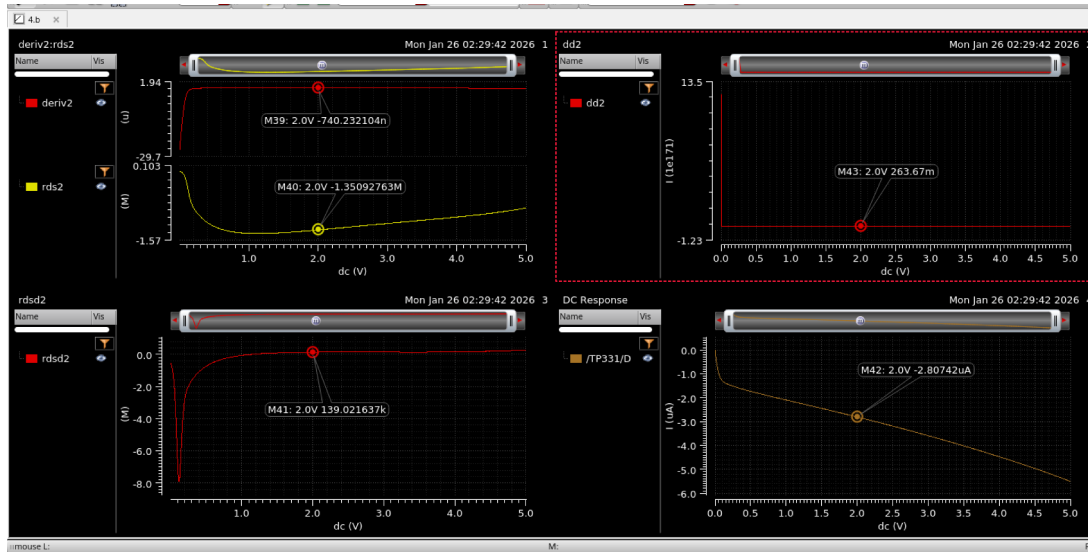
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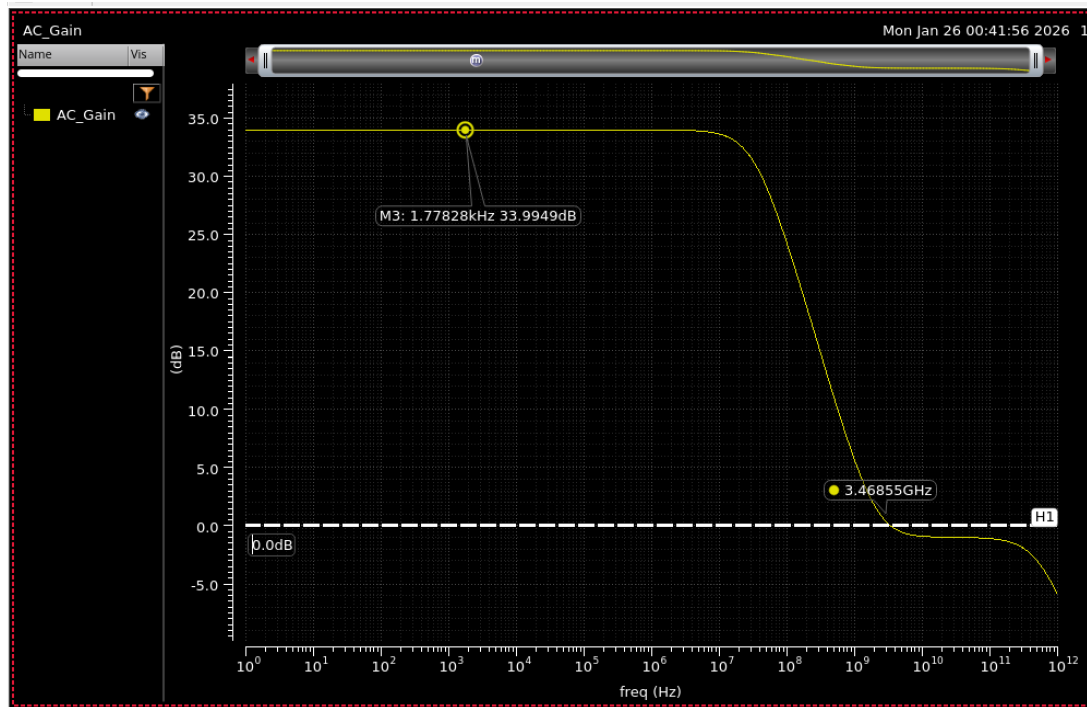


b. Set $V_{GS}=V_{tn}+100\text{mV}$. Sweep V_{DS} from 0 to 5V, plot and gds (dI_D/dV_{DS}) vs. V_{DS} . Plot the output resistance r_{ds} , and its first and second derivative as function of V_{DS} .



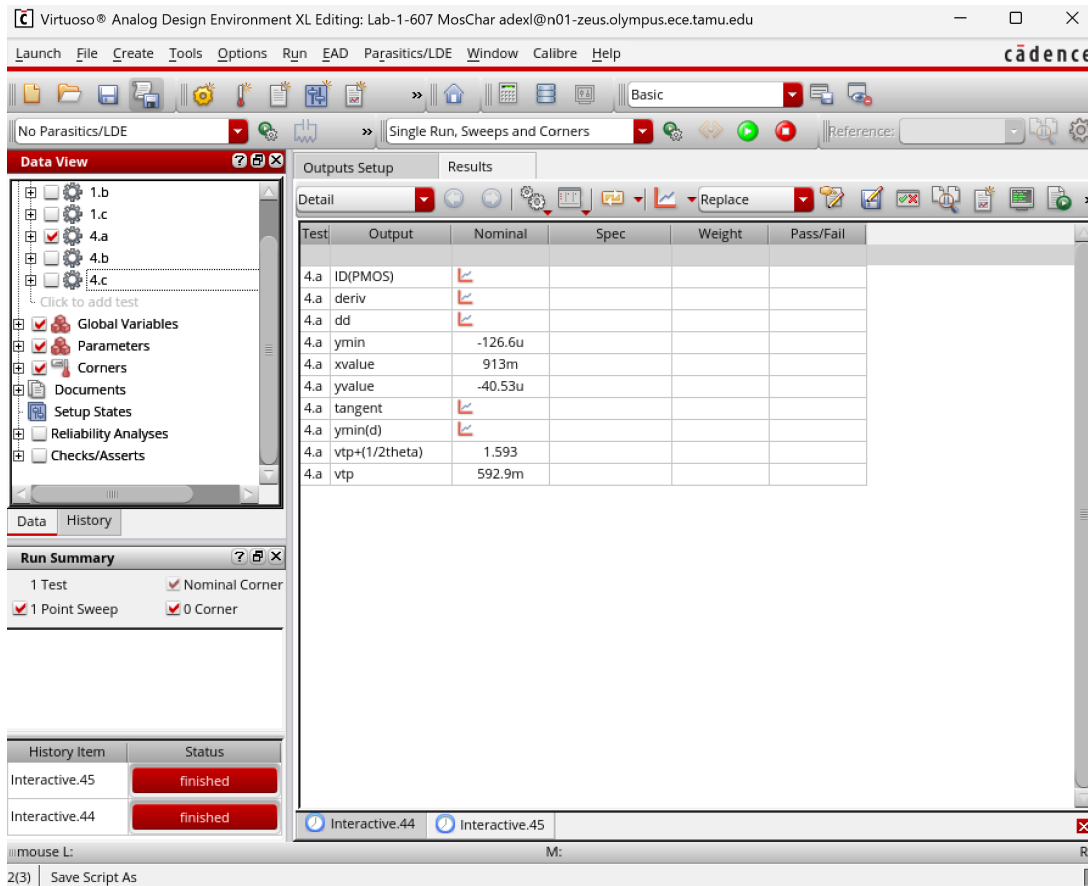
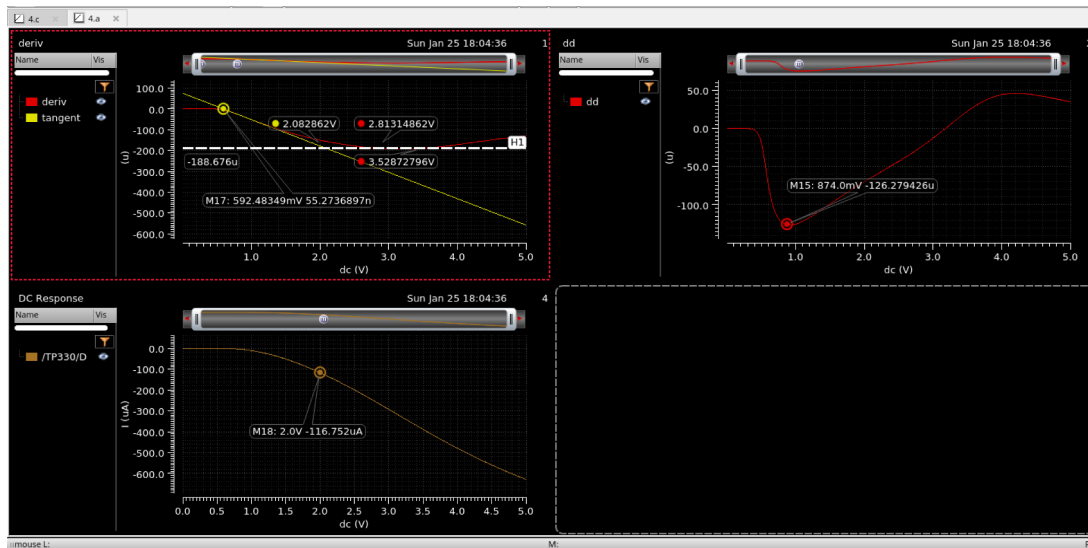


c. Find the intrinsic gain (A_i) and unity-gain frequency (f_t) using AC simulations when $V_{DS}=2V$ and $V_{GS}=V_{tn}+100mV$.

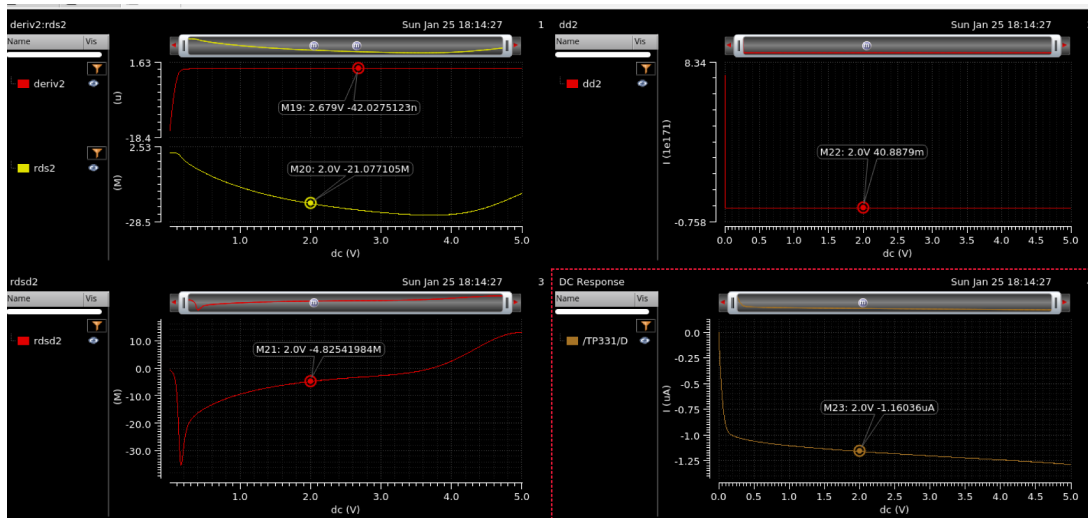


$$3. W/L = 5\mu/1\mu$$

a. Fix $V_{DS}=2V$. Sweep V_{GS} from 0 to 5V, plot I_D and g_m (dI_D/dV_{GS}) vs. V_{GS} , and extract V_{tn} , $\mu_n C_{ox}$ and θ .

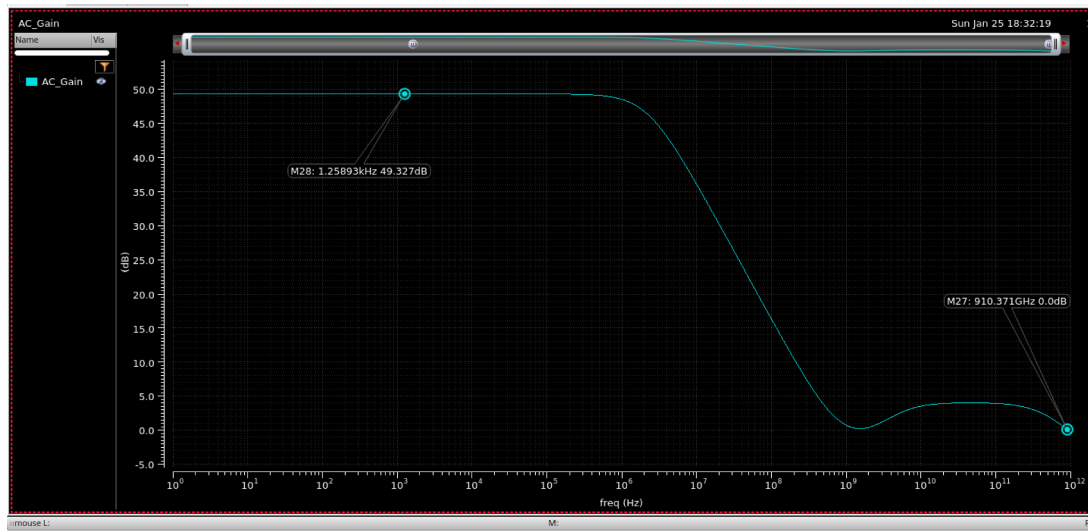


b. Set $V_{GS} = V_{tn} + 100\text{mV}$. Sweep V_{DS} from 0 to 5V, plot and g_{ds} (dI_D/dV_{DS}) vs. V_{DS} . Plot the output resistance r_{ds} , and its first and second derivative as function of V_{DS} .



c. Find the intrinsic gain (A_i) and unity-gain frequency (f_t) using AC simulations when $V_{DS}=2V$ and $V_{GS}=V_{tn}+100mV$.

Positive current:



Texas A&M University ECEN 607: Adv. Analog Circuits & Tech Spring 2026